

TEA: Text Entry Augmentation/ Amplification

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Technical interventions like generative AI tools have become increasingly common in multilingual researchers' workflows, saving time and streamlining complex research tasks. This project examines how such researchers can better engage in equitable research collaboration, information seeking, and knowledge creation in their language of comfort. In particular, we envision a future in which multilingual researchers can read, write, and reason with generative AI tools in their language of choice without any loss in nuance, accuracy, or authority—a future where their language amplifies rather than constrains their ability to think, create, and collaborate. While tools like ChatGPT are widely adopted for literature review, academic writing, and brainstorming, they are predominantly trained and designed for English, reinforcing inequities in scholarly communication. Across our survey and contextual inquiry, we found that these inequities manifest as recurring breakdowns, including mistranslations, uneven explanation depth, and uncertainty about model intent, leading researchers to rely on language-switching and manual rephrasing to preserve meaning. Building on these insights, we designed a low-fidelity prototype enabling bilingual prompt creation and post-generation refinement through highlight-based tools for complexity, length, tone matching, and translation. Qualitative heuristic and simplified user evaluations revealed issues in system feedback, terminology clarity, and feature discoverability, informing our higher fidelity redesign that incorporates tone match, translate, and simplify features. Subsequently, our quantitative evaluation showed improvements in multilingual research workflows, reducing task time and easing cognitive effort. These findings underscore the importance of designing AI text-entry methods for multilingual researchers, enabling more inclusive and equitable AI-assisted research workflows.

CCS Concepts: • **Human-centered computing** → **Interaction techniques**.

Additional Key Words and Phrases: text entry, generative AI, language support

1 INTRODUCTION

Computational tools such as generative AI systems have become increasingly embedded in multilingual researchers' workflows. In particular, researchers across disciplines increasingly rely on generative AI to support research tasks like literature review, paper writing, and brainstorming activities that depend on expressing precise intent via natural language prompts [10, 14]. This project aims to enable such researchers from all linguistic backgrounds to seek equitable support in consuming and producing academic writing. We envision a future in which the researcher's language of choice no longer constrains how they work, but enables precise reasoning, expressive writing, and genuine collaboration with generative AI tools. However, most AI interfaces and models today are developed with English as the default input language, exacerbating existing inequities in scholarly communication in which English dominates publishing and conference discourse [1]. As a result, non-native English speakers often face difficulties in formulating prompts that fully convey their intent, which can lead to less accurate or lower-quality responses and hinder their ability to integrate AI seamlessly into their research workflows compared to native English speakers. Prior work has documented specific obstacles non-native users face when interacting with AI writing assistants (e.g. difficulty interpreting paraphrases without explanations) [8] and noted how platform designs often reinforce monolingual norms [11]. In this project, our promise is to realize a future where multilingual researchers can collaborate, seek information, and create knowledge equitably through generative AI tools, while continuing to work and communicate in the language that feels most natural to them.

While generative AI tools such as ChatGPT are increasingly used to support research by facilitating idea generation, streamlining writing, and enhancing productivity [10, 14], they also

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introduce significant challenges. Users frequently encounter vague or unreliable outputs, including fabricated citations and unclear arguments [10, 14]. For multilingual researchers, these problems are compounded by the English-dominated nature of training data, which creates barriers in prompt formulation and often results in misinterpretations or lower-quality responses [6]. Although recent systems like *Promptor* [12], *PromptPilot* [7], and *Dynamic Prompt Middleware* [5] aim to improve prompt entry, their evaluations remain focused on general or design-oriented contexts rather than academic research. This gap underscores the need to understand the unique obstacles faced by non-native English speakers in research settings.

To address this gap, we first examined how non-native English speakers use generative AI in research contexts. We deployed a survey targeting multilingual researchers, focusing on their language backgrounds, research tasks delegated to AI, usage patterns when interacting with AI in their native languages, pain points in these interactions, and the kinds of language support they would value in the future. Through these questions, we sought to understand how users currently formulate queries, articulate intentions, and navigate challenges in expressing themselves across languages. The survey findings provided a foundation for identifying obstacles and guiding the design of more equitable multilingual interfaces.

Our survey gathered responses from 24 participants across academia and industry, representing native languages such as Hindi, Urdu, Chinese, Korean, Croatian, and Italian, and English proficiency levels ranging from intermediate to near-native. Participants frequently used AI for literature review, academic writing, and translation but rarely interacted in their native languages due to low accuracy, English-first design, and frequent prompt misinterpretation. These findings validated our hypothesis that current AI interfaces lack sufficient multilingual support and underscored the need for designs that enable researchers to express intent in their preferred languages without losing accuracy or efficiency.

Building on these insights, we conducted contextual inquiries with five multilingual researchers to observe how they actually use AI tools for academic reading and writing. These sessions revealed how users switch between languages, verify meanings, and adapt prompts to overcome breakdowns such as mistranslations, uneven explanation depth, and loss of trust. Participants often relied on bilingual prompting, translation cross-checks, and manual rephrasing to ensure conceptual accuracy, showing how language switching serves as both a cognitive aid and a workaround for system limitations. To translate these findings into design opportunities, we developed low-fidelity prototypes, including personas and storyboards to visualize how multilingual researchers' strategies could be better supported through more inclusive text entry mechanisms. These visualizations helped explore design directions that address a subset of our identified user requirements like reducing cognitive fatigue in expressing multilingual research ideas, receiving collaborative AI responses, maintaining control over the conversation, and correcting misinterpretations efficiently during multilingual interactions with AI.

Using insights from our survey, we refined our focus to the challenges multilingual researchers face when prompting LLMs for academic reading and writing, particularly when working with research papers. Building on findings from our contextual inquiries, which revealed both the antecedents and consequences of bilingual text-entry practices, this project centers on designing text-entry interactions that surface users' decision-making, anticipate breakdowns, and mitigate existing limitations.

To address this goal, we created a low-fidelity paper prototype and evaluated it with users and domain experts through heuristic reviews and simplified user testing. This prototype supports non-native English researchers in expressing ideas in their preferred language while co-creating with LLMs. The system allows users to speak their ideas, have them translated and structured into coherent prompts, and edit ambiguous segments before finalizing. After the LLM produces a

response, users can highlight specific sections to adjust tone, complexity, or sentence length, and selectively translate content for better comprehension.

Building on these findings, we transitioned to a functional high-fidelity prototype to evaluate whether our refined interaction techniques could meaningfully improve real research workflows. This prototype allowed us to move beyond identifying usability issues and begin testing the measurable impact of the highlight-based transformation menu on efficiency and cognitive load. To assess this, we conducted a controlled comparison between the prototype and a baseline system without the feature. The following sections detail this evaluation and its implications for supporting multilingual researchers' prompting and reading tasks. By examining objective task performance measures alongside subjective task load ratings, this evaluation helped us determine the extent to which our redesigned interactions address the core user requirements identified in our contextual inquiry and heuristic review phases.

Our findings underscore the importance of designing AI text-entry methods for multilingual researchers, enabling more inclusive and equitable AI-assisted research workflows.

2 RELATED WORK

Generative AI tools have quickly become integrated into diverse fields, ranging from creative writing to professional communication, and their presence in academic research is rapidly expanding. Scholars have reported that tools such as ChatGPT can support researchers and students by facilitating idea generation, streamlining writing, and enhancing productivity [4, 10, 14].

Yet, these benefits are accompanied by significant challenges: users often encounter vague or unreliable outputs, including fabricated citations and unclear arguments [10, 14]. One important factor contributing to these issues is the nature of prompt entry. Kristensson [9] pointed out that improvements in text input systems should not only pursue speed and accuracy but also focus on the "context of use", which are the user's language habits, task goals, and interaction situation. Because most of the training data are in English, non-native speakers may face additional barriers when formulating the prompts, leading to misinterpretations or lower quality responses [3, 6]. Furthermore, multilingual users often "code-mix" between native language and English to compensate for the shortcomings of monolingual input, but existing systems struggle to effectively support this cross-lingual expression [3]. Therefore, text entry becomes a complex cognitive and linguistic process, not just a technical interaction problem.

Recent advances in large language models have led to intelligent prompting systems such as *Promptor* [12] and *PromptPilot* [7], which enable automatic or guided prompt generation for improved text quality. These systems successfully reduce the cognitive burden of prompt engineering, yet their focus remains on technical optimization rather than understanding the user's *context of use*. Specifically, prior work does not address how linguistic background, task goals, and domain-specific constraints shape users' prompt construction behavior in academic settings.

Similarly, recent interface innovations, such as the *Enhanced Text Entry Assistance* framework by Zhang et al. [16], integrate generative AI into mobile keyboards to support content refinement and tone control. While these systems emphasize personalization and cross-application consistency, they primarily assume monolingual English users and overlook multilingual or bilingual interaction patterns. For multilingual researchers who frequently switch between languages or engage in code-mixing, existing solutions fail to accommodate their cognitive and linguistic workflows.

Earlier HCI research has long emphasized the importance of understanding the context of the user through methods such as *contextual inquiry* and *iterative prototyping* [15]. These approaches allow designers to capture the practices of users' in the real-world, identify latent needs, and continuously refine design solutions based on empirical feedback. Previous work has shown that iterative cycles

of sketches, storyboard, and low-fidelity prototyping enable researchers to externalize early ideas and evaluate design concepts before full implementation [2, 13].

In contrast, most recent LLM-based prompting systems still focus primarily on algorithmic performance, providing limited affordances for interactive correction or reflection during text entry. As a result, users often lack mechanisms to control or understand how meaning changes during translation or tone adjustment, reducing interpretability and trust.

Motivated by this gap, our study adopts a user-centered, iterative design approach: we begin with a survey to define our design focus, conduct contextual inquiry to uncover users' cross-lingual prompting behaviors, and then use sketches, storyboards, and paper prototypes to translate user needs into tangible design concepts. Finally, we employ qualitative evaluation methods—including simplified user testing and heuristic evaluation—to identify where and why the design fails to meet user requirements. This methodological sequence reflects established HCI principles that link context understanding with iterative refinement to ensure that system design evolves in alignment with users' linguistic and cognitive contexts.

3 ESTABLISHING FOCUS: INITIAL SURVEY

To explore how non-native English researchers interact with generative AI tools and the challenges they face in text input, we designed a survey questionnaire, tested it in a pilot group, refined it based on feedback, and ultimately released the refined version. The pilot survey questionnaire included demographic questions about age, native language, other fluent languages, and research role. In addition, it included questions about the frequency of generative AI use, the types of models used, the research-related tasks supported by AI, and the challenges participants faced when constructing prompts in different languages.

The main objectives of the survey were to understand the participants' language background, their practical applications of generative AI in research, and their communication strategies when facing prompt-related challenges.

3.1 Method

The pilot survey was particularly designed to ensure that the questions were clearly worded, fit the academic context, and were detailed enough to support the research objectives. The research team brainstormed and developed survey questions around three themes: (1) demographics and language background; (2) frequency of AI use and research applications; and (3) prompt design strategies and communication difficulties. These principles guided the development of the initial questionnaire.

The survey, administered online via Qualtrics, initially required an informed consent statement outlining the study's purpose, duration, confidentiality, and voluntary participation. Participants were required to confirm their consent by checking a box before proceeding. The survey was organized into sequential modules (demographics, AI usage, and text entry challenges) and employed a skip-logic approach.

The initial survey was piloted with a small group of researchers using a think-aloud approach. The inclusion criteria for the pilot participants was - individuals who do research full-time or part-time, are non native English speakers and have used Generative AI in different aspects of their research. Participants read each question aloud and verbally expressed their understanding, while researchers noted any hesitations, misunderstandings, or ambiguities. Feedback from the pilot led to several revisions of the questionnaire:

1. Added explanation: In the announcement at the beginning of the questionnaire, we added an explanation of the "prompt" to avoid confusion among participants.

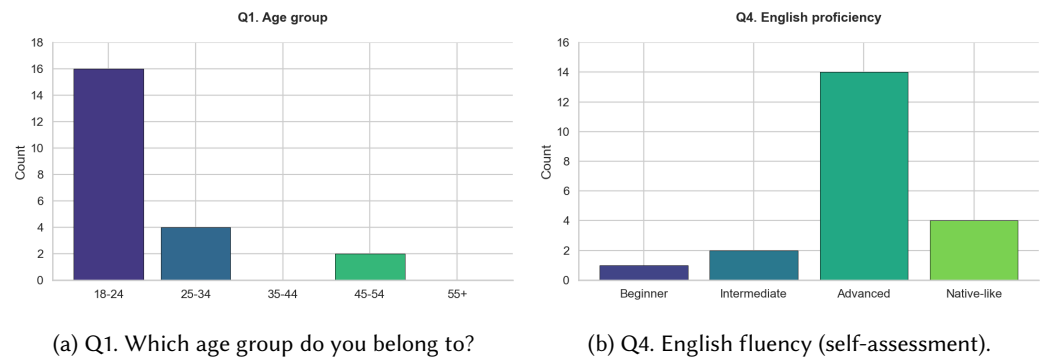


Fig. 1. Participants' demographics and language profile.

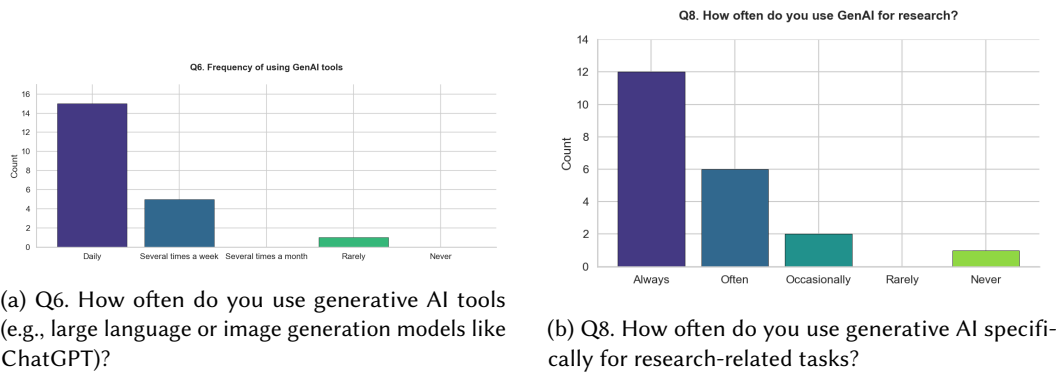


Fig. 2. Participants' overall and research use of GenAI.

2. Refinement: In response to feedback regarding Q3 (which addressed uncertainty about whether multiple languages could be filled in), we revised the question stem to clearly include "you may list all the languages you are fluent in." Furthermore, since not everyone has a research role, we changed Q5 from "research role" to "please state your current occupation" to help respondents better understand the question.

3. Adjustment of options: To avoid overlap or imbalance in options, we added or split some questions. For example, in Q9, "Writing" was split into "academic writing assistance" and "administrative writing assistance." For frequency-related questions, we reassigned the options to prevent overlap.

4. Adjustment of order: The order of questions in the initial version of the questionnaire may have made it difficult to understand. For example, the "Challenges - Solutions - Suggestions" section was broken up, which confused some respondents. Therefore, in the final version, we grouped related questions together to improve logical coherence.

5. Logical Jump Settings: On the Qualtrics platform, we added logical controls to the questionnaire. For example, if the respondent is a native English speaker or has not used generative AI tools, the questionnaire will automatically terminate to ensure that the sample meets the research objectives.

The full text of both the pilot questionnaire and the final revised questionnaire are included in the Appendix (B.1 and B.2, respectively).

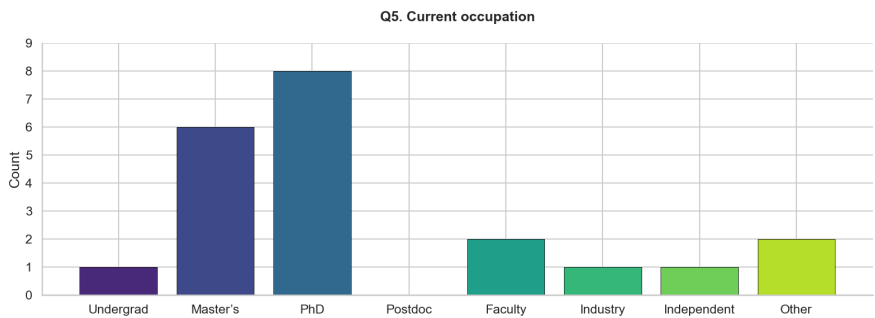


Fig. 3. Q5. Current occupation of survey respondents.

3.2 Tasks and Procedures

Pilot participants were recruited through the group’s personal networks. The primary inclusion criterion was that participants were non-native English speakers with experience using generative AI in academia or research. Every member of the research team participated in the recruitment, resulting in a small but informative pilot sample.

Before beginning the survey, participants received an informed consent statement on the Qualtrics website. This statement explained the purpose of the study, the estimated completion time of approximately five minutes, and guaranteed confidentiality and voluntary participation. Participants were required to check a box to confirm their consent before completing the questionnaire.

The pilot project utilized a think-aloud approach. After completing the questionnaire, participants were asked to read each question aloud and express their thoughts, understanding, and any difficulties they encountered. Researchers closely observed the process, noting any hesitations, misunderstandings, or areas of confusion caused by wording. Because the pilot project was exploratory, focused on identifying potential issues and improving the clarity and structure of the questionnaire, no incentives were offered for participation.

On average, the survey took approximately five minutes to complete, with no section exceeding ten minutes. This short duration was intended to minimize participant fatigue and reduce the likelihood of inattentive responses. To further ensure data quality, the survey allowed participants to skip questions, included an “other” option in multiple-choice questions, and used skip logic in the Qualtrics version to avoid irrelevant responses. For example, participants who identified themselves as native English speakers or had never used generative AI were dropped from the survey early.

3.3 Participants

Before formally distributing the questionnaire, we conducted five pilot interviews to ensure that the survey items were clear and free of ambiguity. The background information of the interviewees is summarized in Table 1.

Table 1. Background information of pilot interview participants

ID	Gender	Highest Degree	Research Experience	English Proficiency	Native Language
P1	Female	Master’s	Yes	Advanced	Chinese
P2	Female	PhD	Yes	Advanced	Macedonian

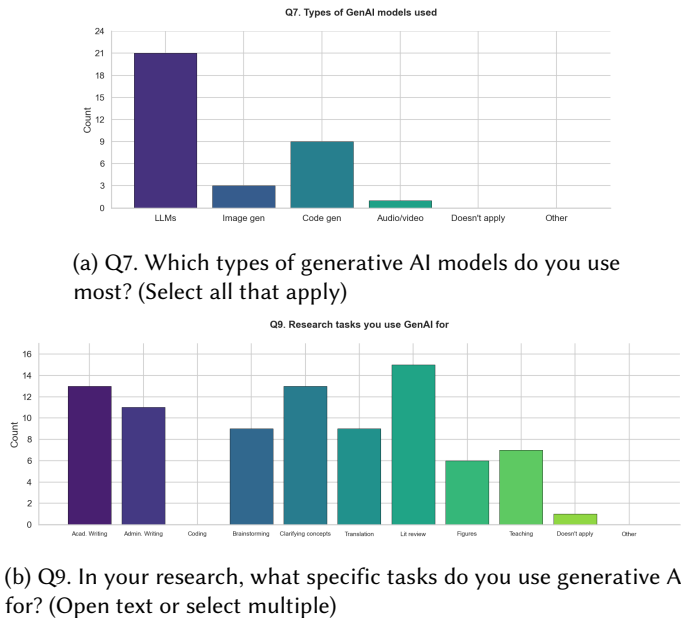


Fig. 4. Survey respondents use of generative AI

We aimed to survey individuals working in research roles across industry and academia. We also only included researchers from non-native English-speaking backgrounds. As seen in Figure 1, a majority of the survey respondents belong in the "18-24" age group, and most self-asses an advanced or native-like proficiency of English. Similarly, most respondents are either pursuing a Master's or PhD degree. Most importantly, a vast majority of the respondents have experience using generative AI tools, either for general use cases or within their research.

3.4 Results

First, from Figure 4a we can see that all survey respondents utilize LLMs, while about half use Code generation tools such as "Copilot" and "Cursor". Moreover, Figure 4b shows that the most common use case for generative AI is in performing literature review, followed by clarifying concepts, and help in academic writing. This helps us narrow our focus on interactions with generative AI focused on these specific sub-tasks instead of any general task. Overall, it also confirms our initial belief that a majority of our focus group of multilingual researchers, are both familiar with generative AI tools and utilize them in their research.

Next, Figure 5a tells us that there is an even split between providing short prompts versus longer more detailed ones. Crucially, Figure 5b accentuates that unclear or incomplete responses by generative AI is a very common reason causing back-and-forth conversations. Here, 5a also points out that a majority of the survey respondents are not aware, or do not utilize more sophisticated prompt techniques that could provide them with better responses.

One of our primary intentions with the survey was to examine whether differences exist in the respondents' interactions with generative AI when communicating in English versus in their native language. First, as seen in Figure 6a, a majority of our survey respondents stated that they either never, or rarely use their native language with generative AI. Our survey also included a follow-up short-response question to this, asking: "If answered rarely or never, can you elaborate

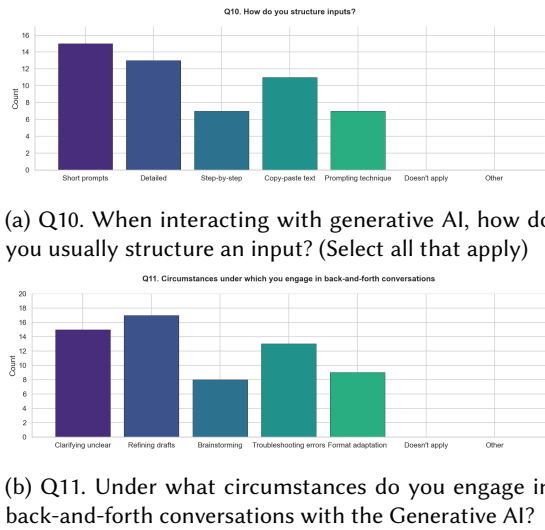


Fig. 5. How users structure input and engage in back-and-forth conversations.

on your choice for Q12?”. Table 2 includes these answers, showing a consistent pattern among the respondents: even when English is not their native language, participants overwhelmingly prefer to use English when interacting with generative AI models. The reasons for this can be broadly categorized into three categories:

1. Work and academic environment. Many respondents emphasized that formal work, research, and higher education are primarily conducted in English, making English a natural choice for AI interaction.
2. Model performance. Some respondents noted that generative AI tools perform better in English than in their native language, with lower error rates and more reliable output. Some respondents also suggested that using their native language could "bias" the results.
3. Habit and cognition. Many respondents described English as their default or instinctive choice for using digital tools, often because these tools are perceived as "English-first." Some respondents stated that they never considered using their native language.

Overall, the reliance on English seems to be driven by a combination of external factors (AI performance, professional norms) and internalized habits (perceiving English as the default option). While some participants acknowledged improvements in native language support (e.g., Hindi), English still dominates in research-oriented environments.

Similarly, Figure 6b and 6c show us that a majority of the survey respondents rarely or never mix their native language with english. Here, it’s interesting to point out that Figure 6d shows a bimodal-like distribution of the number of prompts respondents send to generative AI to reach a satisfactory response relative to English. Since English is one of the most high-resource languages, especially in research and coding, it is expected that generative AI performs better in English compared to lower resource languages, which would in theory correspond to more prompts required in languages different from English. This clearly is not the case in our plot, since respondents with a worse proficiency in English are likely to provide lower quality prompts in English relative to their native language, leading to a secondary peak at "Much less" in Figure 6d. Overall, there are thus two main forces at play here: a) generative AI performs better when used in English, and b) users with worse English proficiency provide better prompts in their native language when compared with English.

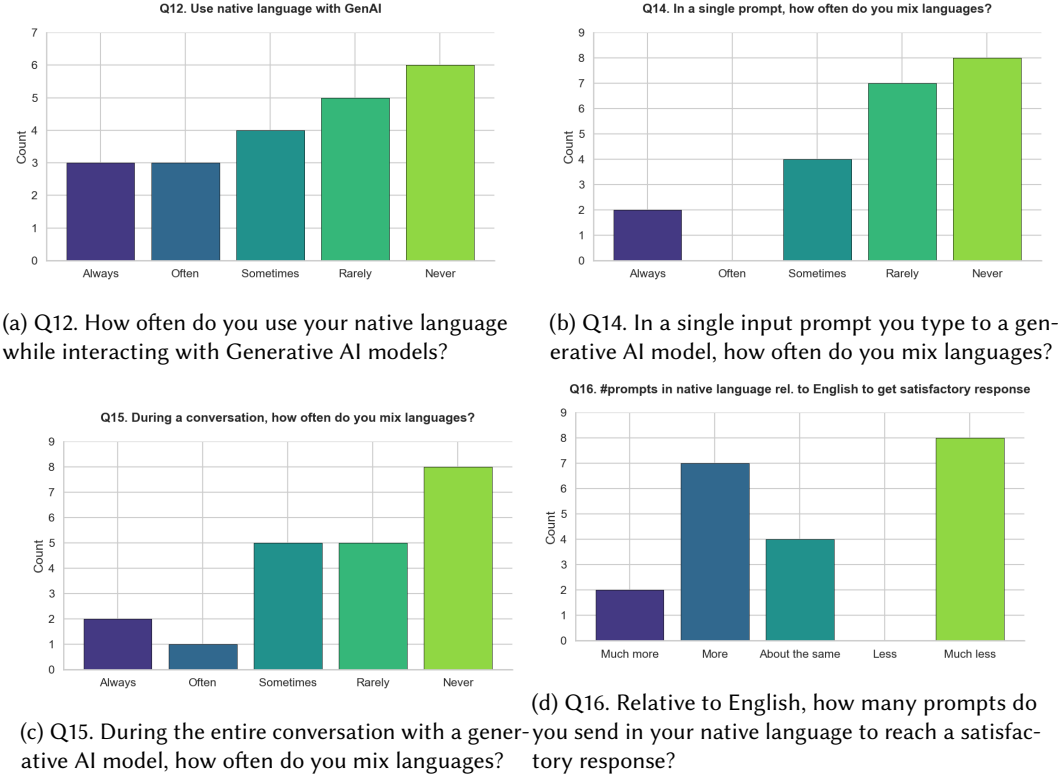
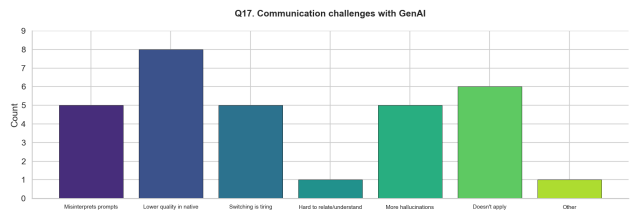
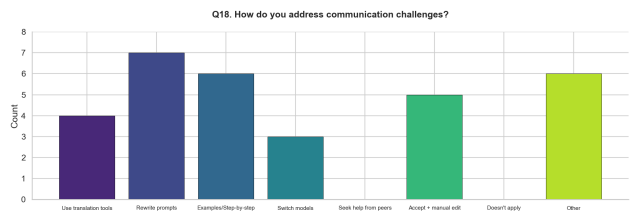


Fig. 6. Use of native language or mix of native language with English.

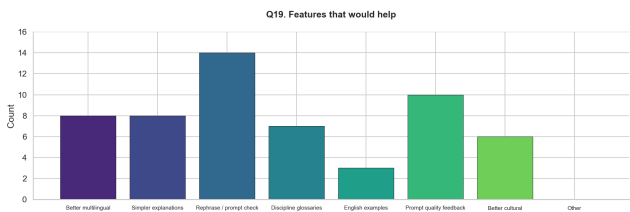
Finally, Figure 7 points out any communication challenges the participants face when using generative AI tools for research tasks, and how they go about addressing these challenges. Here, the most common communication challenge was lower-quality or less relevant responses in the users' native language relative to English. This is directly related to the fact that many languages are lower resource when compared to English, as mentioned earlier. In addressing these issues, the most common approach was rewriting the prompts while providing either more examples or detailed step-by-step instructions. Lastly, the most common feature participants indicated would improve their communication with generative AI was "built-in tools to rephrase or check the clarity of my prompt". Here, it's important to note this is related to the prior mentioned result from Figure 5a, where most users are not familiar with more sophisticated LLM prompting techniques. This is clear since many such methods are part of recent AI research, and thus largely inaccessible to most users.



(a) Q17. What communication challenges do you face when using generative AI tools for research tasks in your native / non-english languages (compared to English)? (Select all that apply)



(b) Q18. If you mentioned any challenges in Q17, how do you usually address or work around them? (Select all that apply)



(c) Q19. What features or changes in generative AI interfaces would make it easier for you to communicate effectively? (Select all that apply)

Fig. 7. Communication challenges with generative AI and suggested features for improvement

Table 2. Open-text responses to Q12 follow-up (“If answered rarely or never, can you elaborate on your choice?”)

Responses	
Most of the work I need it for is in English.	
Most of what I deal with is English content, hence it is in some sense more “harmonious” to keep the prompts in English too.	
The models I use aren’t good in my native language.	
I’ve always approached western tools as being English-first, so my instinct is always to use English. Additionally, all formal work like academic research, education in my country, etc. is primarily in English. This makes English the default choice for such tasks, while everyday communication and most casual communication happens in my native language.	
I’m used to communicating in English for work and don’t feel the need to use Hindi.	
ChatGPT now has a good knowledge of Hindi and is improving, but still there is room for improvement. I don’t want to hamper my outcome in order to give feedback to ChatGPT on Hindi. Also, for research English is always preferred for communication.	
Mostly operate in English.	
Models work better with English than my native language (Korean).	
Never work in Hindi, as most of the code and analysis is in English and I am very proficient in the same.	
I always talk in English with LLMs and never in my native language (Hindi).	
It never occurred to me to use my native language. I do try to clarify English concepts many times—to simpler English terms.	

4 UNDERSTANDING CONTEXT OF USE: CONTEXTUAL INQUIRY

4.1 Method

In our contextual inquiry, the purpose was to understand how multilingual researchers interact with generative AI tools during academic writing, particularly how their linguistic background shapes their text entry behavior, prompt construction, and response interpretation. The study was conducted in participants' natural work environments, such as university offices or home study spaces, where they typically write research papers or prepare academic materials. Observing participants in these authentic contexts allowed us to capture real-world challenges and adaptations rather than simulated behaviors.

Each session was designed to document how participants formulate and enter prompts, focusing on the breakdowns, adaptations, and strategies that occur when switching between their native language and English. During the sessions, the interviewer encouraged participants to “teach us how you usually complete this task” to elicit natural, self-directed workflows.

With participants' consent, audio, video, and screen sharing were used to record the process, capturing their typing sequences, language switching patterns, and system feedback. The recordings were later coded into event chains and visualized through sequence diagrams, flow maps, and affinity diagrams to identify recurring patterns and pain points.

4.2 Tasks and Procedures

Before the session, all participants were informed of the study purpose and procedure, and verbal as well as written consent was obtained for recording and data use. The interviewer will let them know it voluntary and that recordings would remain confidential and de-identified. The typical session lasted 30–120 minutes.

Each session started with a short chat to get everyone ready and get to know each other, then we had a look around the environment. This involved the participants explaining or showing how they had set up their workspace (for example, the lighting, how they had arranged their devices and how they worked together). Then, they were asked to do an academic writing task like making a paragraph better, summarising a paper, translating text, or creating instructions. They could use ChatGPT or other AI tools for this. Depending on their usual practice, they were asked to explain what they were doing and why while interacting with the system.

During the task, the interviewer observed prompt formulation, language switching, editing behaviors, and breakdown moments (e.g., accidental early submission, mistranslation, system misunderstanding). Participants occasionally shared artifacts such as previous AI sessions, screenshots, or drafts to illustrate their workflow. No prior prompt templates or examples were provided to avoid bias.

After the activity, a short debrief invited reflection on perceived challenges and desired interface improvements—for example, automatic keyword suggestions, better handling of large PDFs, or improved bilingual output. We were committed to quality control, so all recordings were reviewed straight away after each session. We checked our observation notes against screen captures and transcriptions to make sure the data sources were consistent. The next step was to code and triangulate the patterns across the five participants.

4.3 Participants

We recruited 5 non-native English researchers who routinely use generative AI for *text-entry* in academic work, with the explicit goal of contrasting intermediate versus fluent bilingual practices in authentic contexts of use. Participants were included if they (i) self-identified as non-native English users, (ii) engaged with ChatGPT or similar systems for writing-related tasks, and (iii) could

demonstrate their typical workflow on their own devices. This ensured ecological validity and surfaced language–tool tensions central to our focus on text entry. We decided on five participants because similar interaction patterns and text-entry challenges began to repeat, suggesting that we had reached qualitative saturation for our research focus.

Participants were recruited by convenience / peer sampling in graduate and professional researcher communities. Prior to each session, the interviewer explained study purpose, confidentiality, recording plans, and voluntariness; verbal (and where applicable written) consent for audio/video/screen recording was obtained. Sessions were conducted either remotely (Zoom, participants’ usual workspace) or in-person, preserving natural work settings. No monetary incentives were offered.

The sample captured disciplinary and linguistic diversity aligned with our research questions: a data/survey science graduate student with intermediate English proficiency who scaffolds prompts in Chinese but inserts English technical terms; a humanities graduate student fluent in Korean and English who relies on bilingual summarization/translation and contends with large-PDF limits; and a fluent Hindi–English software developer who ideates in Hinglish but types in English due to keyboard/script frictions. In all participants, AI was used several times a week for paragraph polishing, literature summarizing, translation, tone control, and prompt iteration.

Each contextual inquiry lasted approximately 30–120 minutes, including warm-up and debriefing. We targeted five sessions to reach pattern saturation for a focused qualitative study of text entry breakdowns.

Table 3. Participant demographics and study context

ID	Gender	Native Lang.	Eng. Level	Discipline / Role	AI Tasks (Text Entry)	Mode	Time
P1	F	Chinese	Intermediate (7/10)	Grad student (Data Sci)	Paragraph polishing	Zoom	~50m
P2	F	Korean	Fluent	Humanities grad student	Summarization, translation, tone control; screen-shot workaround	In-person	~45m
P3	M	Hindi	Fluent bilingual	Software dev / researcher	Idea drafting, literature check, Hinglish→English typing	Zoom	~45m
P4	F	Macedonian	Fluent	Professor	Summarization, translation	Google Meet	~50m
P5	F	Chinese	Fluent	Master Student	Paper reading	In-person	~120m

Participants varied in their language proficiency and task focus (Table 3). Intermediate users (P1) relied on Chinese frameworks to clarify intent; fluent bilinguals (P2, P3, P4, P5) strategically switched languages to manage efficiency and nuance. Across all cases, all participants used ChatGPT as their primary writing assistant and experienced similar issues with translation accuracy, oversimplified tone, and a lack of clear explanations for changes.

4.4 Results

4.4.1 User Goals and Existing Behaviors.

- Summarization Structuring Knowledge** Participants use ChatGPT to condense and organize academic material into clear, structured summaries that align with their teaching or research goals. They often rely on the tool to translate and reframe complex English research into another language while maintaining conceptual accuracy. In doing so, they aim to simplify difficult content for lectures, projects, or learning new subjects. However, users want more control over how summaries are generated. They seek options to adjust structure, tone, and depth so the output fits academic expectations rather than generic text. Many participants also expressed the need for assistance during summarization, such as automatic keyword or terminology suggestions that reflect their research field. This indicates a preference for tools that not only summarize but also support comprehension and learning. For participants reading in new or unfamiliar domains, summarization serves as a way to build familiarity with recurring concepts, methods, and language. Overall, this category reflects how researchers are beginning to use ChatGPT as a cognitive partner that helps them structure knowledge, internalize ideas, and communicate complex material across languages and disciplines.
- Writing Revision Support** Participants rely on ChatGPT as a writing partner to refine, simplify, and rephrase their academic or professional text. They share their drafts and use the system to improve grammar, flow, and clarity while maintaining their original ideas. The tool is often used to simplify dense sentences, paraphrase complex arguments, and elevate the tone of writing to meet formal or academic standards. In several cases, participants extended this use beyond academic writing, asking ChatGPT to help with emails or general communication tasks. For bilingual users, the tool became a practical aid for switching between languages depending on the purpose of the task. Some preferred drafting in their native language, such as Chinese, to express ideas easily, then translating or reworking the output in English to refine terminology and accuracy. Overall, this theme reveals that users do not see ChatGPT as a replacement for their own writing but as an adaptive collaborator that supports them through multiple stages of revision. It helps users save time, reduce language-related barriers, and build confidence in producing polished and comprehensible written work.
- Project Context Motivation** Participants' engagement with ChatGPT is deeply shaped by their immediate academic or professional context. Many are new to their research projects and turn to the tool as a way to build foundational understanding of the key concepts, methods, and terminology they will repeatedly encounter. For these users, ChatGPT serves as a bridge between initial exposure and confident participation in their project work. Motivation often stems from external deadlines or mentorship structures, such as upcoming check-ins or task reviews with supervisors. Participants use ChatGPT to prepare for these moments, ensuring they can demonstrate progress and comprehension. The tool also supports early-stage exploration when users are refining broad or unclear research ideas. Across cases, participants combine self-directed learning with model-assisted brainstorming, often shifting between in-person discussions, supervisor guidance, and GPT-supported exploration in Google Docs. This shows how ChatGPT functions as part of a larger learning ecosystem where users develop ideas iteratively and independently while staying aligned with project expectations and academic
- Understanding and Learning through Native Language Support** Participants use their native language as a key strategy for deepening comprehension when engaging with complex research materials. They describe being able to understand difficult concepts with greater ease and less cognitive strain when explanations are provided in their first language.

ChatGPT plays an important role in this process by translating, paraphrasing, or summarizing material into familiar linguistic and cultural frames that make academic content more approachable. Several participants alternate between languages as a form of cognitive cross-checking, using bilingual translation to verify meaning and ensure conceptual accuracy. This practice supports confidence not only in their understanding of material but also in their ability to discuss it with mentors and peers. For many, native language translation functions as both an accessibility feature and a learning aid that reduces anxiety about language barriers. Participants also use ChatGPT to preview or summarize English research papers in their native language before reading them in full. This approach provides a mental outline that enhances efficiency and comprehension. Overall, this category highlights how bilingual and multilingual researchers leverage generative AI as a bridge between understanding and articulation, using their native language to anchor learning and build confidence in academic communication.

4.4.2 *Existing Interaction Workflows with Generative AI Tools.*

- **User initiates conversation with LLM** Participants begin by opening ChatGPT on their laptop, which automatically starts a new chat. Previous chats are often ignored, as they are not relevant to the immediate task. Logging in can be a small friction point, since users are not always signed in and sometimes share accounts. One participant, for example, needed to call their husband to retrieve the updated password. Users often switch between devices like iPads and laptops, finding this combination useful but not completely seamless.
- **User chooses language of interaction** Language choice is a conscious process. Many participants select English even when their inner thoughts or discussions are in another language, such as Hindi or Chinese, because the ChatGPT interface feels more “suited” to English and Western contexts. When writing prompts, some users switch between their native language and English to refine meaning, correct typos, or confirm translations. Others use translators to make their messages more precise. Participants often type an initial idea in English to verify it before refining it in their native language or vice versa. When seeking conceptual explanations, users sometimes copy terms directly from papers and ask ChatGPT for short, precise definitions.
- **User crafts a prompt** Prompt creation is an intentional and iterative process. Some users adopt templates learned from online courses, showing a more structured prompting style. They carefully revise and polish their inputs, often pausing to recheck tone or phrasing before sending. Others, however, do not put much thought into the process and rely on habitual or spontaneous phrasing. The level of maturity in prompting tends to vary as some have developed refined habits, while others approach it casually based on immediate needs.
- **User evaluates model responses** Once the model produces output, users engage in a layered evaluation process. They start by assessing the overall coherence and relevance, then move to tone, logic, and substance. The review process resembles academic reading, where users compare AI-generated content with actual papers. Some participants read line by line, vocalizing and moving the cursor to maintain focus.
- **Interaction breakdown workarounds** When interactions fail or confusion arises, users employ several strategies to restore understanding. They rephrase prompts in English to avoid mistranslations or incorrect substitutions by AI. In some cases, participants clarify their ideas through follow-up questions or additional prompts when initial responses feel incomplete. Others rely on practical fixes. For example, inserting English keywords into multilingual prompts to maintain consistency. These adaptive behaviors show how users work around friction points to keep the communication flow intact.

4.4.3 Accomplishments in Existing Interactions with Generative AI Tools.

- **Obtain satisfactory results** The participants expressed satisfaction with the quality of the output and the research outcomes from interacting with the AI. The participants were provided with more complete and detailed responses and references, and the development of their research drafts was advanced. Furthermore, participants reported that the AI's optimisations in language structure and expression logic rendered the results of English input clearer and more organised.
- **More easily understand the content in the native language** Participants found that when complex concepts were explained in their native language, they were able to gain deeper insights with less effort and demonstrate greater confidence in academic communication. It was also noted by some participants that, while English provided more accurate model output, communicating in their native language allowed them to think more naturally and feel more connected.
- **Improved Efficiency and Positive Experience** Participants found that obtaining explanations and definitions through ChatGPT was more concise and efficient than searching websites or blogs, reducing the stress of information overload. Furthermore, the model's immediate responses were found to foster positive emotions, thereby helping users to maintain greater focus during research and writing. Upon satisfaction with the results, the findings were incorporated directly into their formal documents. This suggests that interaction with Gen AI has the potential to enhance research efficiency and the user experience.

4.4.4 Challenges in Existing Interactions with Generative AI Tools.

As we can see in C.6, there are 8 different categories of challenges in the existing interactions users have with Generative AI. We can then split these categories among 2 broad challenge themes, one dealing with model-side responses and their qualities, and second theme concerned with user-side behaviors, their interactions, and adaptations.

Model-side challenges.

- **LLMs hallucinate** A crucial limitation of LLMs that cause multiple breakdowns for our users interviewed during the contextual inquiry is the fact that LLMs, especially when conversing in low resource languages, often hallucinate. In particular, the LLM may invent and use terminology that does not exist, confusing the user. Moreover, in especially low resource languages like Macedonian, the LLM often incorrectly translates English into Bulgarian or Serbian instead.
- **Time to obtain response is too long** Another limitation in the current interactions is that ChatGPT's new reasoning capability often takes a long time before outputting a response. In addition to the default breakdown of additional time the user must now spend waiting, there is another breakdown as the user has no indication of how long the "thinking" process will take.
- **Output length/explanation does not match user requirements** A very common limitation we observed during our contextual interview is that the output length and explanation level are not compatible with users prompt requirements. In particular, either the LLM output is too long and over-explained, or on the contrary too short and under-explained. In both cases, this causes more back-and-forth with the LLM, as the user must now reclarify their requirements.
- **LLM output is in an inconsistent format** Compared to the previous category, we also more broadly observe that the LLM's output is not consistent with the user prompt requirements. In particular, certain users felt that the LLM's output was unnatural and different from

their everyday interactions. Moreover, this is connected to another common breakdown, where the LLM provides information that is irrelevant to the user.

User-side challenges.

- **User mistakes/slips and language setup disrupt LLM connection** From our contextual inquiry, we observed that the way the LLM handles a user mistake/slip disrupts the users connection to the LLM. In particular, a user pressed "send" too early, not having fully finished their response. The user then paused the model's reasoning, corrected the prompt, and sent it back the LLM. However, this slip led to the LLM misunderstanding the user's prompt. Moreover, language setup issues, involving having to use a different native language script, or constant switching between native and English often disrupt the connection to the LLM.
- **User quits** The most severe breakdown we observed is when a user receives many dissatisfactory responses and simply gives up on the interaction. In one cases, this was because the user felt the LLM was no longer as effective as themselves in completing the task. In another scenario, the user simply received too many LLM errors in a row, resulting in their frustration, stopping the interaction.
- **User adapts prompting strategy** A common category of interaction challenges was one where users had to change and adapt their prompting strategy in order to avoid LLM misunderstanding. For instance, one user changed from prompting in their native language to only prompting in English, believing it would make the prompt requirements more clear to the LLM. Another user, who had a goal of obtaining a 10 sentence summary of an English research paper in their native language, first asked the LLM to give a 10 sentence summary in English, followed by translation of those 10 sentences to the native language, believing that the LLM is simply better at summarization in English. All interpretations in this category accentuate the point that LLMs perform better in English when compared to the user's native language, especially in cases of low resource languages.
- **User adjusts workflow due to LLM limitations** Similar to the previous category, in this one we observe users that must adapt their whole workflow to overcome an LLM limitation. For example, one user pointed out that after an update to ChatGPT, they've had to include way more details in their prompt to receive the same quality of output when compared to before. Moreover, another user felt that the LLMs perform worse when given a whole PDF document, having to instead send the document broken up into multiple screenshots.

5 USER REQUIREMENTS AND FUNCTIONAL CONSTRAINTS

Updated R1. Users must be able to access research-related information that is factually accurate and aligns with their stated research goals.

Grounded in context: Multiple nodes show users relying on GPT for "information seeking," "summarization," and "concept clarification." Breakdowns occur when "accuracy of generated responses" and "hallucinations" appear.

Updated R2. Users must be able to check AI-generated information through built-in verification steps that are accessible from within their task workflow.

Grounded in context: Based on "User doubts the accuracy of translation or key concepts," "wants verification through manual cross-checking," and "takes screenshots of readings and uploads to GPT instead of an entire PDF."

Updated R3. Users must be able to communicate in English and in their native language such that English research-related conversations convey the same meaning and information as equivalent non-English conversations in real-life research contexts.

Grounded in context: From "switches back to English for clarity and smooth conversation," "user

self-re-prompts in native language to correct GPT,” and “builds ideas through bilingual brainstorming.”

Updated R4. Users must be able to obtain responses that maintain equivalent formality, politeness, familiarity, and factual accuracy across all languages used in interaction.

Grounded in context: “Receives a response in native language which feels unfamiliar,” “responses in English feel too formal,” and “alignment and familiarity in conversation is important.”

Updated R5. The system must provide indicators that show how a user’s language choice influences the model’s tone, detail, or reasoning.

Grounded in context: Nodes “model misunderstood requirements” and “user needs to rephrase their idea to bridge gap between task efficiency and expression of thoughts” show uncertainty about GPT’s reasoning across languages.

Updated R6. Users must be able to identify whether a GPT response is supported by evidence or external references before using it for academic reasoning.

Grounded in context: “User doubts the AI’s understanding of their native language,” “believes GPT is hallucinating key concepts,” and “wants verification before applying results.”

Updated R7. The system must allow users to edit or update previous inputs and must generate new responses that reflect those changes.

Grounded in context: “Clarifies idea through follow-up turns in conversation” and “model response is not satisfactory → user closes ChatGPT task.”

Updated R8. Users must be able to identify when GPT’s response does not match the academic tone and conceptual depth required by the task.

Grounded in context: Supported by “feels ChatGPT is not up to the task,” “responses feel too formal,” and “mood response is not satisfactory.”

Updated R9. The system must allow users to make successive revisions to generated content through supported actions such as edit, undo, redo, and saving/restoring progress.

Grounded in context: Nodes “writes down a version of the idea in Google Docs,” “uses AI to refine written idea,” and “builds upon idea through in-person brainstorming.” Supported by evaluation feedback indicating a need for user control, freedom, and the ability to revisit earlier stages without restarting the workflow.

Updated R10. Users must be able to easily revisit, track, and reuse successful prompts or conversation histories within TEA.

Grounded in context: “Takes screenshots of readings and uploads to GPT later,” “reuses earlier prompts for summarization,” and “calls back previous conversation threads.” Strengthened by evaluation findings on visibility of system status and user feedback, which revealed a need for better progress tracking and version recall.

Updated R11. Users should switch between different [styles of writing for research] more efficiently and with decreased cognitive load.

Grounded in context: “Mental model forms in one’s chosen language,” “mental load increases when switching languages,” and “alignment and familiarity help sustain focus.” Enhanced based on evaluation findings that unclear labels and limited system feedback increased mental strain for multilingual users.

Updated R12. Users should switch between languages when discussing research related topics more efficiently and with decreased cognitive load.

Grounded in context: “Confidence in prompting in native language,” “language helps in comfortable rational communication,” and “user mentally translates ideas to verify meaning.” Revised based on evaluation findings showing confusion over terminology such as “complexity” and “ambiguity detected,” which limited clear conceptual expression.

Updated R13. Users must receive responses that avoid evaluative judgments and clarify the user’s input through concrete suggestions.

Grounded in context: “Responses feel robotic or too formal,” “user feels unsatisfied and closes GPT,” and “alignment and familiarity in tone improves engagement.”

Updated R14. Users must be able to reject or discard AI-generated output and continue their task without losing existing work or context.

Grounded in context: “User wants to save time and refine later,” “switches back to manual brainstorming,” and “user closes GPT feeling unsatisfied.”

Updated R15. Users must be able to identify and correct GPT misinterpretations without needing to rewrite their full request from scratch.

Grounded in context: “Model misunderstood requirements,” “user needs to rephrase idea to bridge gap,” and “prior experiences of limited model notions within non-native language.”

Updated R16. The system must provide feedback that shows how specific elements of a user’s prompt affect GPT’s reasoning and outputs.

Grounded in context: “User tests different prompt structures,” “notices differences in response when changing phrasing,” and “prompt changes influence result structure.”

Updated R17. Users must be able to identify, verify, and correct hallucinated or mistranslated content.

Grounded in context: “User believes GPT is hallucinating key concepts,” “verifies translation manually,” and “user self-re-prompts to correct GPT.”

6 INITIAL DESIGN AND LOW FIDELITY PROTOTYPES

After completing the first round of individual design sketches, our team conducted a design critique using the “I like, I wish, What if...” method (Table 4). Each member presented their ideas and received feedback on usability, user experience, and technical feasibility. During this process, we realized many initial concepts overlapped, particularly in bilingual interaction, voice input, and prompt optimization. Therefore, we decided to consolidate our respective strengths and integrate the most promising features into a unified design.

Based on the review outcomes, we divided the final concept into two primary design scenarios, each representing distinct phases of the user workflow:

1. Before sending prompts—Voice-based multilingual text input and translation
2. After receiving output—Optimized and simplified interface based on highlighted content

Table 4. Summary of Design Critique Using the *I like, I wish, What if* Framework

Design Author	Feedback Summary (Reviewer 1–4)
Marko’s Design	Reviewer 1: “I really like the idea of sliders. We can definitely build such sliders based on common language breakdowns like complexity, tone, or formality. It’s almost like an over-the-counter filter for multilingual users to fine-tune their text. Framed more explicitly around text entry, it could act as an adaptive interface layer that dynamically adjusts linguistic parameters in real time, allowing users to both express intent and control model behavior. What if we could also make these controls customizable, so users can select which linguistic dimensions, such as clarity, emotion, or technicality, they want to manage?” Reviewer 2: “Figure1: I really like the way you designed the sliding adjustment for complexity and length — it’s a very creative idea! I’m curious, though, about how the “tone match” on the left works. As a user, I might not be entirely sure how to interact with it. Storyboard1: The story is very engaging! I love the translation feature, and I also think the Chinese writing is really well done. Storyboard2: Great story! From this storyboard, I got a clearer idea of how the tone match function works. However, I’m still a bit unsure can users select the tone, or do they input it directly? Maybe having a selection option could help reduce users’ cognitive effort.” Reviewer 3: “Love the idea. I know this is a low fidelity prototype however I think we can add a couple toggles to choose a language because i dont see that feature right now. I think this is very thorough and covers almost everything for the user requirements. If we were to choose one person’s features I think this would be a good way.” Reviewer 4: “I like that this design nailed many key points surfaced from contextual inquiries; specifically, sliders make the text entry experience a lot less cognitively demanding by mitigating many menial useless efforts spent entering text. I wish that there could be more fine-grained scales that we can use to adjust academic-related things, too (e.g., have people generate their own sliders in their native language). What if we try offering users abilities to make custom scales in their native language?”

Design Author	Feedback Summary (Reviewer 1–4)
Marko’s Design	Reviewer 1: “This is great for trust and source verification! I think it might be a bit generic, so we could tailor it more toward research and multilingualism. For example, framing it around how multilingual researchers navigate linguistic ambiguity, translation accuracy, and disciplinary tone in text entry could make it more grounded in their real writing practices.” Reviewer 2: “Figure2: It is very nice! I really like how you use “confidence” and “source” to show the AI’s level of certainty in its output! I’m wondering, though, does this box appear only when the user clicks on the text, or does it stay visible all the time? If it appears upon interaction, I think that would make the interface look especially clean and elegant! Storyboard3: I really like the setting of this storyboard, the window and the trees make the scene feel very lively and realistic.” Reviewer 3: “I think this is good for not just vonfidence scale for the Prompt and response but the users response since tha twas a main behavioural concern for most interviews that we saw.” Reviewer 4: “I think that this design nailed many key points surfaced from contextual inquiries. Specifically, sliders make the text entry experience a lot less cognitively demanding by mitigating many menial useless efforts spent entering text...”

Design Author	Feedback Summary (Reviewer 1–4)
Ishika’s Design	Reviewer 1: “I love this design, I think it’s great to have it split up into 3 parts, all relevant to the user and their interaction with the LLM. My only question is how does the third panel: prompt generated, work? Is that generated by an LLM? If so, how can we make sure that generation itself does not suffer from some of our breakdowns” Reviewer 2: “Sketch1: I really like this design! I think incorporating voice input is very valuable. I have two thoughts. First, since voice input often enables real-time conversation with AI, perhaps we could add a button or toggle that lets users easily switch into this “translation + prompt” mode. Second, I find the idea of highlighting or marking users’ reviews very interesting. I think it could connect well with Marko’s design — for example, if a user is not satisfied with a prompt, maybe they could adjust it directly through that interface. Storyboard1: This story is very detailed! I really like how the whole process is clearly described. I think if we could include some information about the challenges our personas faced and explain why certain design choices were made — it might make the overall result even more convincing! ” Reviewer 3: “I like how you added the environmental context separately than the storyboard. I also like the steps that you add, that would be nice to have cognitively. also love the idea of voice to text. thats a feature that we should add. My only concern was that we studied mostly text entry for our contextual interviews than voice to text.” Reviewer 4: “I like the use of voice input. It is really interesting and can get rid of the current issues faced by multilingual researchers in getting succinct, good explanations about things they see in English papers. Voice makes it intuitive and perhaps is a better way of intuitively entering questions. I wish that we can make it clear to the symbol what the voice input is referring to. What if we introduce a feature that lets people point to and select text as they enter voice input?”

Design Author	Feedback Summary (Reviewer 1–4)
Nina’s Design	Reviewer 1: “Love the highlight interaction, it’s quite interesting and intuitive. The only contention is around how we could actually implement this in a technically feasible way, especially without interrupting the user’s writing flow. Still, I really like this idea a lot. What if we extend the highlight mechanism to not just act locally but also translate the identified aspect, like tone, clarity, or confidence, into general model responses? This could help apply similar adjustments across the text, making the overall interaction more adaptive.” Reviewer 2: “Storyboard1: I really love the style of this storyboard – it feels just like a comic strip! The storyline flows very smoothly too. Sketch: This is a very interesting sketch! I think using highlights to show question points in real time is really helpful. If a small pop-up window could appear next to the highlighted text to allow users to ask questions immediately, it might make the interaction even more convenient.” Reviewer 3: “This was very thorough and written deeply. I like the separation by color. At first I was a bit overwhelmed by the colors and a tad bit confused but as I read I got the concept more. I also like the separation of things.” Reviewer 4: “I really like the highlighting idea, where Cheryl can highlight specific parts of the paper she’s confused by, and how it’s separated in different colors. I also love that she gets explanations in both languages side by side, which would likely reduce the cognitive load from otherwise scrolling through different responses. My only critique would how cases in which Cheryl might want a more mixed response that contains both English and Chinese would be handled.”

Design Author	Feedback Summary (Reviewer 1–4)
Xinyu’s Design	Reviewer 1: “I love the many ideas you’ve explored. I have a special affinity toward the keyboard interaction - I would genuinely love to use it myself. I wonder if we could make it more research-oriented by integrating some of these ideas into existing research tools. For instance, what if features like adaptive sliders, multilingual tone adjustment, or citation-aware suggestions were embedded directly within writing or annotation interfaces that researchers already use? It could transform text entry from a purely mechanical task into a more reflective and context-aware research activity.” Reviewer 2: “could be prototyped with more guidelines since it is so complex I was a bit lost without guidance. But I think you have a lot here in terms of user requirement coverage and featurisms. I love how you thought of toggling between buttons. Thats something we don’t see.” Reviewer 3: “I really like the use of controls. Reading papers in English can already be very challenging for non-native speakers... having hot key-esque features can reduce many pain points during text entry that are now (at least according to people in contextua; interviews) anticipated. I wish that people could tell the system what symbols they want. What if we introduce actual hot keys that people can define?” Reviewer 4: “I really love the idea with the buttons and something I’ve never thought of before. It seems like a way more efficient way of toggling through languages instead of having using mouse/track pad, and something we can easily add to any design. The one thing I was confused about is how toggling the languages works after the prompt is generated, does it change it in place, or does it generate new ones?”

Design Author	Feedback Summary (Reviewer 1–4)
Sabrina’s Design	Reviewer 1: “Nice idea. I like the source reference bit, especially how it shows the translation alongside the corresponding English source. The main question in my mind is how we could implement this without disrupting the writing flow or making the interface too complex. This might also be more useful for users who are not very proficient in English, though it could still add value in maintaining accuracy and trust for all users.” Reviewer 2: “I really like your storyboard, the drawings are beautiful! From what I understood in the sketch, does it mean that users can directly annotate the content after sending the PDF to the LLM? Want to make sure I understood that correctly.” Reviewer 3: “I really like this idea of being able for GPT to directly show what concept is being communicated. It addresses requirements related to making this 3 way communication (human, ai, paper) easier. I think that it would nice if people could save further energy from having text entry suggest for them via hover/grey text. What if when people select something, the system attempts to suggest things people could look into?” Reviewer 4: “I love how the prompt and pdf are split up side by side where Donna can simultaneously view her prompt and highlighted parts of the pdf. I also love the storyboard, both how well it depicts Donna’s day, and how the design addresses her needs. The one part I was confused by are some of the specific details of how Donna interacts with the design, is the pdf essentially displayed on the right, where she can highlight any parts she’s confused by? Also, I would have liked to see more details on how some of the multilingual aspects were addressed — can Donna choose or interact with which concepts get translated in English versus Burmese?”

6.1 Personas

Emma Laangs Persona:Emma Laang (check 29) was created as a consolidated persona that combines insights from international students, researchers, and educators into one cohesive representation. She reflects the shared needs for confidence, accuracy, and trust when using GPT tools across languages and academic work.

We included elements such as her bilingual background, teaching role, and tendency to cross-check information to capture users’ real struggles with language switching, verification, and overreliance on technology. These features ensure the persona covers core user requirements including accuracy and factual reliability, preservation of tone and language, transparency in AI responses, and reduced cognitive load through efficient workflows. Emma represents the wider emotional and practical journey of users who depend on GPT to produce clear and authentic work while striving to feel capable and confident.

6.2 Sketches

Our final design is displayed in Figure 30, with a more detailed prototype in Figure 34. After our design critique, we proceeded to finalize our design by combining features from individual designs

that altogether address many of our prior specified user requirements. At first, we wanted to find some combination that can address every single user requirement, but quickly realized that it would not be feasible and having too many features could potentially overwhelm Emma. Moreover, we realized that our individual designs can be grouped into ones focusing on interactions that happen during the initial prompt stage, and ones following the generation in which Emma interacts with the LLM output.

With this in mind, we decided to combine Ishika’s design, displayed in Figure 27, with Marko’s design, displayed in Figure 28, in addition to changing those existing designs based on specific design critiques. After reviewing Ishika’s design, we recognized that voice inputs could enable users to communicate with AI tools more naturally, allowing them to express their thoughts directly in their native language without losing context or nuance. We also found that enabling users to upload or add text content and layer verbal instructions on top could help overcome interaction barriers that make prompting in native languages less efficient.

However, this design alone does not meet the need for personalizing AI-generated responses in terms of language, tone, and complexity, as highlighted in the contextual inquiry. It is equally important to make such customizations simple and intuitive for non-native English speakers, since our findings show that adjusting linguistic and stylistic aspects can be particularly challenging for these users.

For our post prompt interactions, our main features are the following:

- (1) The user can toggle the complexity and length of any highlighted text
- (2) The user can translate any highlighted text into their selected language of choice
- (3) The user can provide either a description, or their own writing for ChatGPT to perform a tone match, and rewrite selected text using similar tone and sentence structure to the user’s requirement
- (4) The above 3 features work directly over ChatGPT’s output. That is, whenever the user performs any of the above 3 features, the new text is changed on top of the old text, and no new prompts are created.

In particular, following the critiques to Marko’s design, we kept the complexity and length sliders identical to their original design based on reviewer satisfaction during the design critique. Our primary change was to the tone match feature, displayed in detail in Figure 34. In particular, the prototype illustrates how specific text containing wording used in physics, Emma’s research area, can be used to rewrite the highlighted text Emma is confused by to use terminology and phrasing that Emma can better understand. Moreover, one concept we made more clear in our refined designs and final prototype is that complexity and length are orthogonal concepts. In particular, changes to the length should not affect the complexity of the words, and similarly, changes in complexity should not affect the length. These design components are illustrated in the bottom half of 34, where we can see how the length of the highlighted text remains identical even as the complexity slider is changed.

Finally, the above combination of both pre-prompt and post-prompt features address a majority of our existing user requirements. Check 30 and 31 to refer to our final sketch ideation to merge the designs and visualize the environment of interaction. These are also added below.

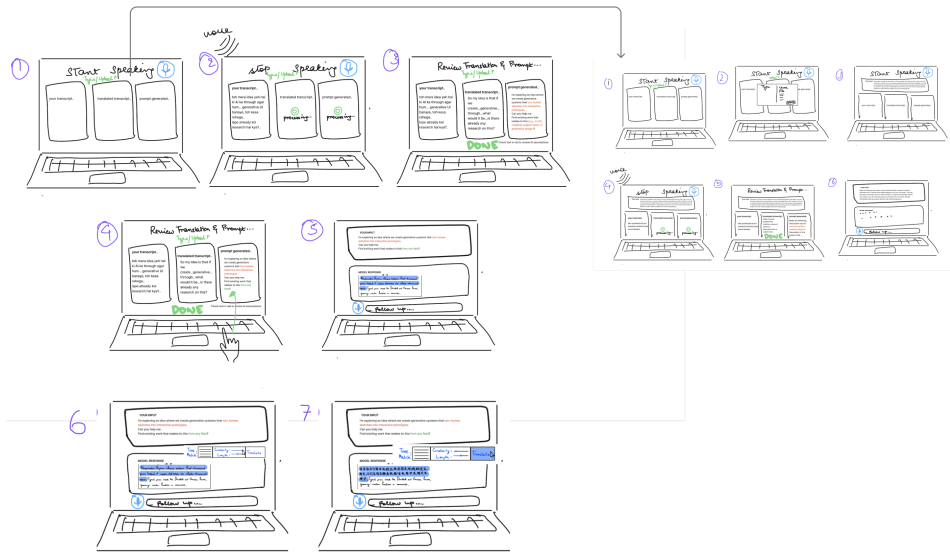


Fig. 8. Voice text entry and bilingual prompt generation.

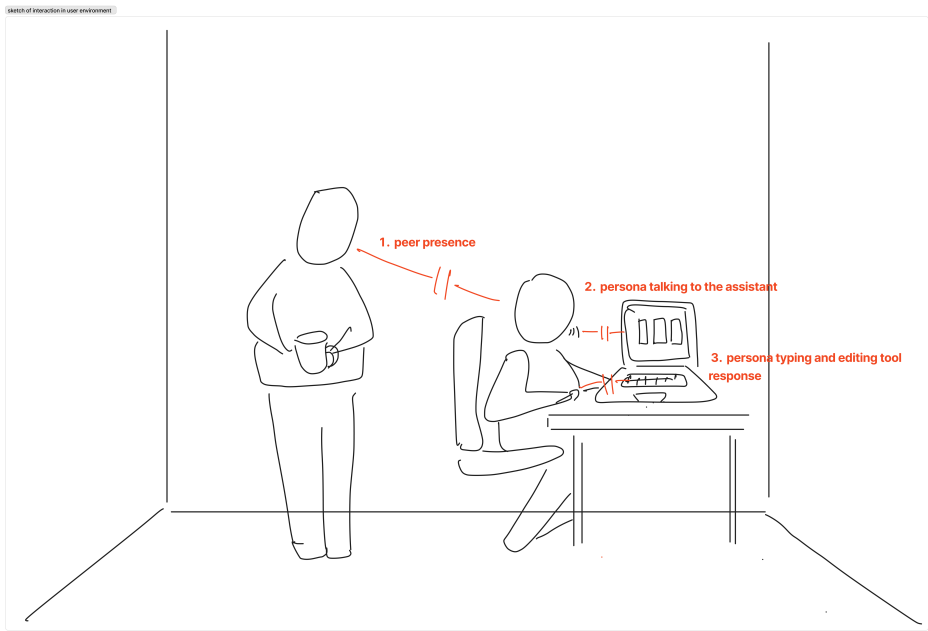


Fig. 9. Sketch showing how the interaction might be realized in the user's environment.

6.3 Storyboards

In the first storyboard (Figure 10), Emma, a non-native English-speaking researcher, is preparing to discuss her thesis ideas with her advisor. After a long day of work, she feels exhausted and lacks the energy to type in English.

Emma discovers a new design. This feature allows her to speak in her native language, with the system automatically translating it into English and generating professional prompts. If the generated prompts don't meet her expectations, she can upload writing guidelines for refinement. The system then combines her voice input with the uploaded guidelines to produce a more accurate version tailored to her needs.

During design critiques, we identified voice input as one of the most critical features for non-native speakers. Group feedback highlighted its advantages in reducing cognitive load and enabling natural multilingual expression.

This storyboard demonstrates how the design helps Emma overcome these obstacles by simplifying idea generation and language translation.

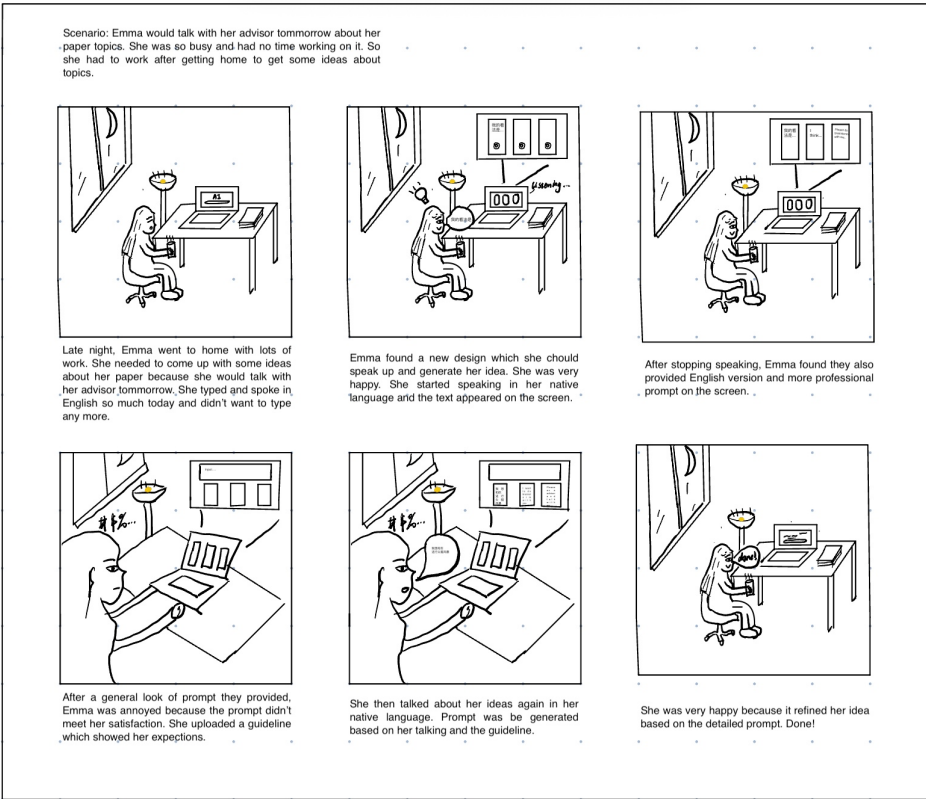


Fig. 10. Storyboard 1: Voice Text Entry and Bilingual Prompt Generation.

The second storyboard (Figure 11) depicts Emma studying complex English concepts for her doctoral thesis defense. As a non-native speaker, she struggles with specialized terminology. To address this, we introduced an interactive highlighting and modification system. She can select difficult paragraphs, triggering an instant pop-up window where she adjusts tone, complexity, and length via sliders. She can also translate the text into her native language with one click to deepen comprehension.

Reviewers appreciated the interactive transparency design philosophy, empowering users to visualize and control the complexity of AI-generated text rather than passively accepting it.

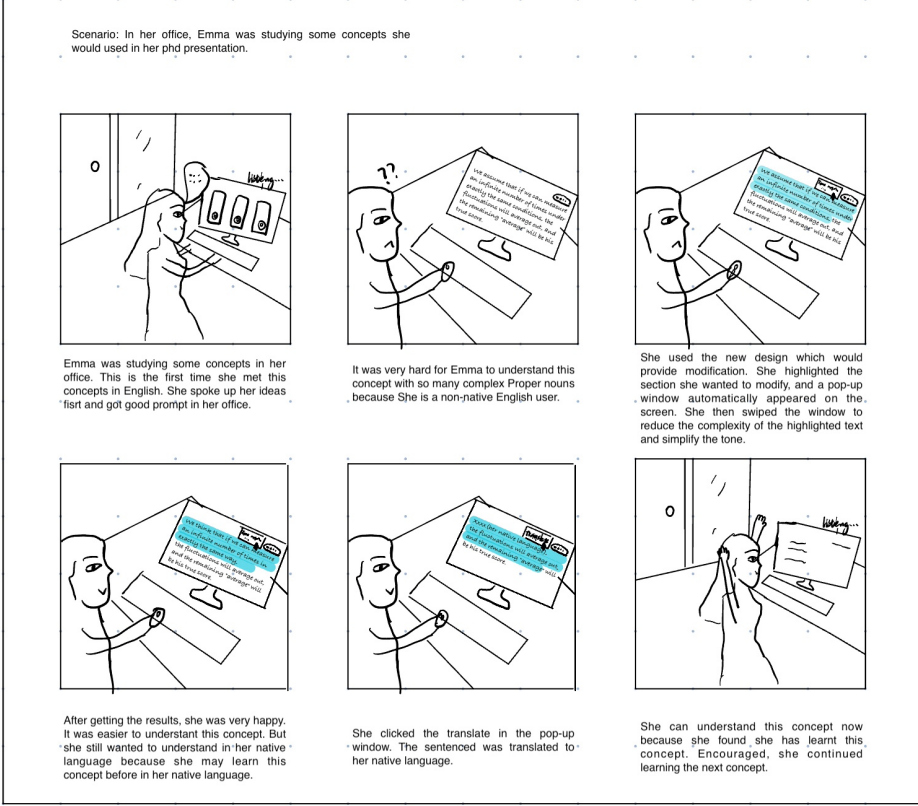


Fig. 11. Storyboard 2: Highlight-based refinement interface.

6.4 Paper Prototype

Our final paper prototype, shown in Figure 34, gives a detailed description of how Emma might interact with our design. We envision Emma to initiate her interaction with the tool (tentatively called TEA) through the interface. If Emma chooses to explain her thoughts through voice, she can click on the *Speak your mind* option. This takes her to the screen where she can click on the record option. On clicking the option, Emma can talk about her idea as she sees it being transcribed. She can then review the generated translations and prompts and edit any discrepancies. She can also go to the *Add Text* option and add additional inputs or texts to be added to the voice instructions. Once Emma is happy with the prompt, she can click on *Done* and move to the screen that generated the response to her prompt. Once the response is generated, Emma could also perform actions to modify the generated response and personalise them to her requirements. The bottom half of Figure 34 clearly displays all steps Emma might take to better understand the highlighted piece of writing. Our first sequence displays the function of the complexity and length sliders, where we first decrease the complexity, and afterwards increase the length. Next, our prototype illustrates the "Tone match" feature, where our design rewrites Emma's highlighted text using physics terminology obtained from the inputted physics paragraph. Finally, our prototype illustrates the translate feature, converting the highlighted text from English into Burmese, Emma's native language.

There are several ways one could "Wizard of Oz" this prototype. The simplest and easiest to implement method would be a laptop made of paper, where one flips through possible paper screens in the order shown in Figure 34. Alternatively, a higher fidelity prototype could be one where the user interacts with an LLM, which in disguise is some human that can change the text according to our prototype's features.

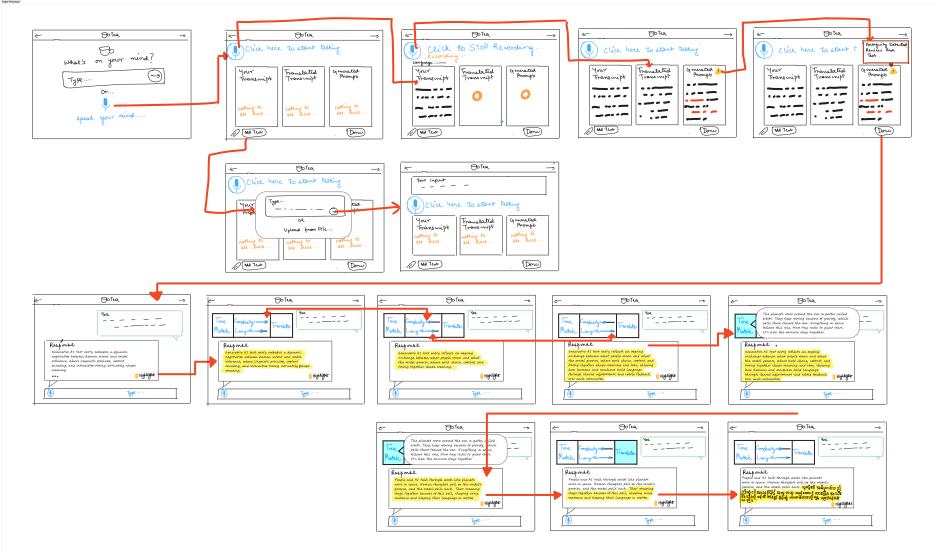


Fig. 12. Final paper prototype

7 USABILITY EVALUATION

7.1 Heuristic Evaluation

7.1.1 Method. The heuristic evaluation was a key part of our design process, helping us understand how well our prototype, TEA (Text Entry Augmentation and Amplification), supports multilingual researchers who read, write, and summarize research papers in different languages. The goal was to assess whether TEA's interface felt intuitive to users with experience in human-computer interaction and to determine how well it followed established usability principles.

Each team member conducted one heuristic evaluation session with an assigned HCI expert. These experts were selected from within our course and had backgrounds in human-computer interaction, usability evaluation, and interface design. Working with these experts allowed us to collect technical feedback that connected design intentions with practical usability concerns.

The evaluation followed Nielsen's Ten Usability Heuristics, which include: visibility of system status, match between the system and the real world, user control and freedom, consistency and standards, error prevention, recognition rather than recall, flexibility and efficiency of use, aesthetic and minimalist design, help users recognize and recover from errors, and help and documentation.

Each evaluator identified usability issues, linked them to one or more heuristics, and rated their severity on a scale from zero to four, where zero meant no usability problem and four meant a critical issue that required immediate attention.

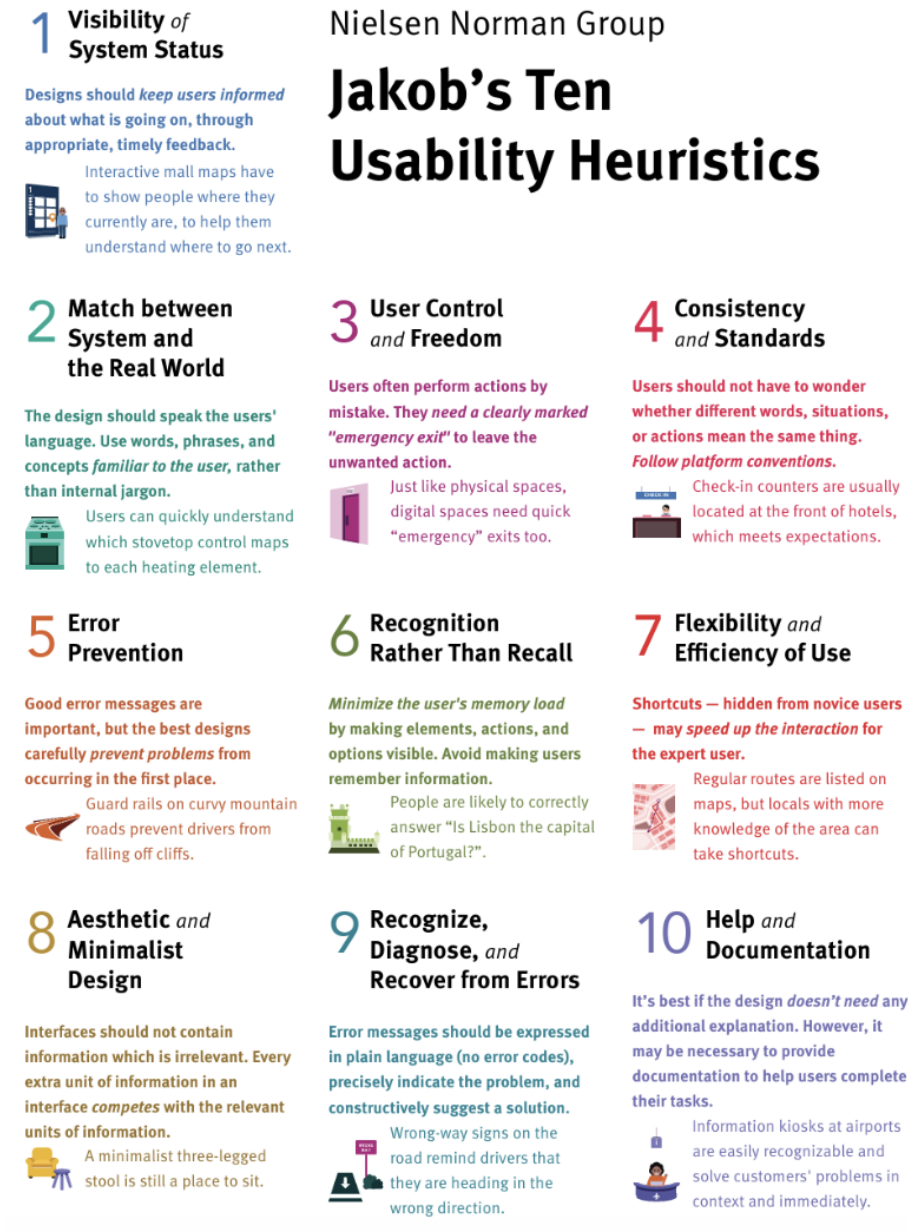


Fig. 13. Nielsen's Ten Usability Heuristics

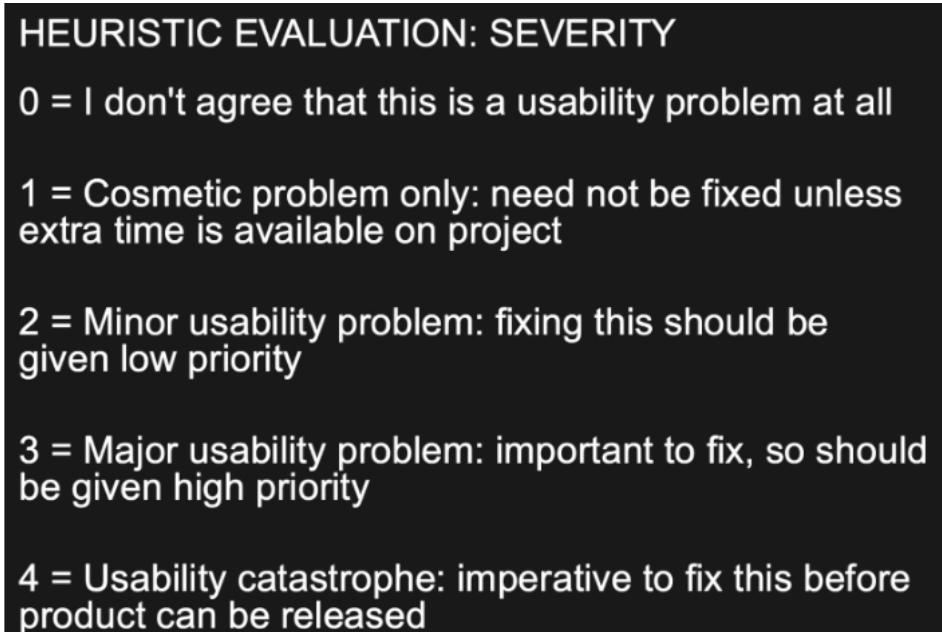


Fig. 14. Heuristic Severity Rating Scale (0–4)

7.1.2 Tasks and Procedures. All evaluators followed the same standardized protocol to maintain consistency across sessions. The evaluations were conducted using the same low-fidelity paper prototype, shown in Figure 1. This prototype was printed and placed on top of a laptop screen, allowing facilitators to simulate interaction using a Wizard of Oz method. The facilitator manually flipped panels, wrote on paper cards, and narrated transitions such as recording, translation, and summary generation to replicate the system’s behavior.

Each session began with an introduction explaining the purpose of TEA and its goal of helping multilingual researchers create short, presentation-ready summaries of academic papers. The facilitator then demonstrated the main task, showing how a user would speak or type their ideas, view them as a transcript, see a translation, and generate a summary that could be refined through the Tone Match, Complexity, Length, and Translate features.

Throughout the demonstration, the evaluator was encouraged to think aloud, ask questions, and share observations about confusing or unclear elements. After the walkthrough, each evaluator reviewed the design using Nielsen’s Ten Heuristics, noted any issues, and provided a severity rating with suggestions for improvement.

Each evaluation lasted approximately thirty minutes. To ensure quality and consistency, every facilitator used the same script and sequence of actions. Evaluators reviewed the same features, workflow, and task scenario, which allowed our team to later combine results without differences caused by varying procedures.

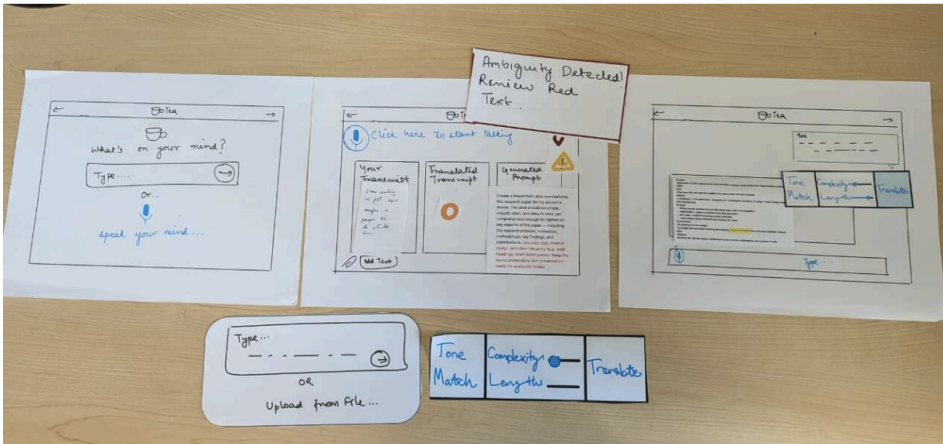


Fig. 15. Paper prototype used in heuristic evaluations

7.1.3 Participants. A total of five HCI experts participated in the heuristic evaluation, one for each team member's prototype. These participants were advanced graduate students or researchers enrolled in the course with training in usability evaluation and interface design.

- Tejas Maire evaluated Sabrina May's prototype and has experience in HCI evaluation and usability testing.
- Anthony Liao evaluated Xinyu Lin's prototype and focuses on design consistency and feedback mechanisms.
- Anika Misra evaluated Ishika Joshi's prototype and has experience in multilingual usability and documentation.
- Tomas Garcia Lavanchy evaluated Marko Veljanovski's prototype and studies user control and AI-assisted design systems.
- The fifth evaluator, a CSE PhD student, provided feedback on the remaining group prototype and has prior experience with user-centered design in computational research contexts.

All evaluations were conducted in person during class or in nearby study spaces. Each expert gave verbal consent to participate after being briefed on the purpose and method of the evaluation. Since these were course-based peer evaluations, no incentives were provided, and consent was informal but clearly obtained before each session began.

The number of participants was decided based on the Discount Usability Testing principle (Nielsen, 1994), which recommends using five evaluators to capture the majority of usability issues during low-fidelity testing.

7.1.4 Results. After completing five heuristic evaluations, our team compiled the findings into one collective report. Each member contributed written notes, severity ratings, and evaluator quotes. We organized all issues by the Nielsen heuristic each one violated and merged duplicate findings into shared categories so that trends could be compared across sessions.

Overall, the evaluators agreed that TEA has a strong and relevant concept for multilingual research collaboration, yet several interface issues reduced clarity, feedback, and user control. Across all sessions we identified twenty one unique usability issues with an average severity of 1.8 out of 4. Of these issues, two were rated as critical, six as major, and the remainder were minor.

The most frequent theme involved visibility of system status. Participants could not always tell when the system was recording, translating, or generating a summary, which led to hesitation and

confusion. Evaluators recommended progress indicators, brief loading text such as “Generating summary,” and clear visual activity cues. Figure 16 shows an example panel referenced during discussion.

Table 5. Combined Heuristic Evaluation Matrix Across All HCI Experts

Usability Problem (Heuristic #; Avg Severity)	E1	E2	E3	E4	E5
No visible feedback while recording or summarizing (#1; 3)	3		3	2	3
Unclear terminology such as “complexity” and “ambiguity detected” (#2; 3)	3	3	2	3	3
No undo, redo, or back option during workflow (#3; 4)	3	3	4	3	4
Inconsistent button or label wording (#4; 2)	2	2		3	2
No upload confirmation or error prevention check (#5; 2)	2			2	2
Hidden controls such as Tone Match and Translate (#6; 2)	2	3		2	2
Limited customization or tone presets (#7; 2)	2			3	2
Visual clutter when multiple sliders are shown (#8; 1–2)	1		2		2
No clear retry option for microphone or connectivity issues (#9; 3)	3		3	3	2
Missing tooltips, help icons, or examples (#10; 2–3)	2		3	2	3

Heuristic Evaluation Notes and Reports

Notes	
Heuristic	Issues
#1: Visibility of system status	Slider interactions lack preview of changes before committing
#2: Match between system and the real world	The complexity feature is ambiguous. Not many systems offer this. Users may not be familiar with this concept. No indication difference between complexity/length
#3: User control and freedom	No undo/redo functionality visible after applying adjustments. Cannot correct voice recognition errors easily
#4: Consistency and standards	Users may be confused about when to use sliders
#5: Error prevention	Unclear scope of ambiguity highlighting.
#6: Recognition rather than recall	Controls not always visible in users' field of vision
#7: Flexibility and efficiency of use	DJ-ing the controls may get tedious.
#8: Aesthetic and minimalist design	Too many controls for some users
#9: Help users recognize, diagnose, and recover from errors	No audio quality assurance.
#10: Help and documentation	Unclear what "complexity" means practically

Nielsen's 10 Usability Heuristics and Issues Identified

Report			
Usability Problem	Heuristics	E1	Recommendations
Some controls may not be intuitive to invoke nor easily accessible	#8; 4; 6; 7	3	Allow for more collapsible controls; offer keyboard shortcuts.
Unclear and ambiguous behavior of complexity slider	#2; 4; 10	4	Provide examples on hover; make complexity clearer with "before/after" examples.
No undo when using complexity sliders	#1; 3	3	Show a real-time preview of text changes as sliders are moved and add arrows with keyboard shortcuts for the sliders
Confusing behavior of ambiguity highlighting	#5	2	Allow users to dismiss or flag highlights and help the system learn.
No documentation for how to be aware of and address audio transcription issues	#3, 9	2	Make uncertainty in transcription more clear with squiggly lines.
Identified Usability Problems, Heuristics Violated, and Evaluator 1's judgement on severity and recommendations			

Fig. 16. Prototype section where evaluators identified missing system feedback

Language clarity was another recurring theme. Several evaluators noted that terms like “complexity,” “prompt,” and “ambiguity detected” were difficult for non native English speakers to interpret and increased cognitive effort. They recommended simpler wording, concise definitions, and supporting visuals to make the interface more approachable.

The lack of user control and freedom was also identified as a critical limitation. Users could not undo actions or navigate backward once they advanced through the workflow, which discouraged exploration and made error recovery difficult. All five evaluators mentioned this limitation, and most rated it as severe. Addressing this requires the addition of undo and redo as well as clear back navigation.

Additional issues of moderate severity included missing confirmation for uploads, hidden or non intuitive controls, and visual clutter when several sliders appeared at once. Evaluators also emphasized the need for helpful onboarding elements such as tooltips, example prompts, and short explanations of each feature. Figure 17 shows the slider controls most frequently discussed during evaluation.

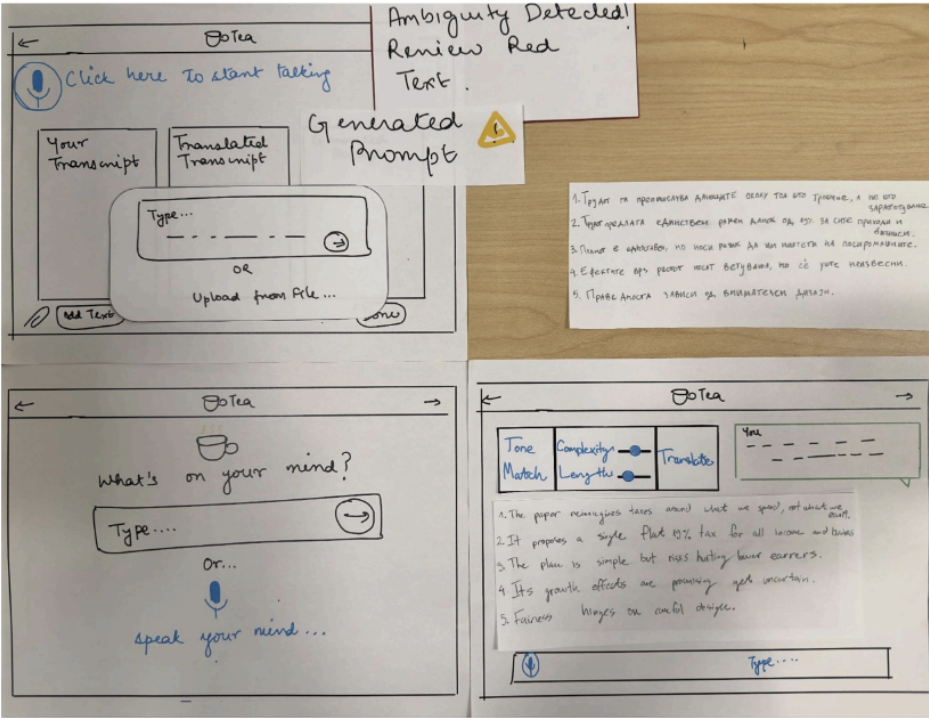


Fig. 17. Slider controls for Tone Match, Complexity, Length, and Translate used during evaluations

In summary, the combined heuristic evaluation shows that TEA is conceptually strong but requires clearer system feedback, simplified terminology, undo and redo functionality, and accessible help. These findings directly informed our next iteration plan and guided refinements to user requirements so that the prototype can evolve into a more transparent and supportive tool for multilingual researchers.

7.2 Simplified User Testing

7.2.1 Method. The purpose of simplified user testing is to evaluate how useful and usable our designs are given a certain set of user requirements. Hence, we conducted the simplified user testing hoping to gain a better understanding of how our target stakeholder group interacts with

our prototype, observing any issues as they come up so that we can improve and quickly iterate upon our design.

Simplified user testing is primarily used since it is a fast and cheap way to evaluate a design against specific user requirements by having users interact with the prototype. Usually, it tends to find the most severe issues that come up in a design, and thus it is extremely useful during the early stages of design when we want to quickly iterate. A crucial component of simplified user testing is the think-aloud approach, wherein the user must think out loud while they interact with the prototype. This helps us as the evaluator to interpret issues the user faces with the prototype. For simplified user testing, we decided to test our prototype against the following 2 qualitative user requirements:

- (1) R4: Users must receive consistent tone and conceptual accuracy across languages
- (2) R8: Users must be able to identify when GPT's response does not match the academic tone and conceptual depth required by the task.

Based on these 2 requirements and our understanding of the context of use, we devised the following user goal to be given to the participant: Obtain a short bullet-point summary of a research paper to be used for an upcoming presentation.

7.2.2 Tasks and Procedures. The specific steps we followed in the simplified user testing are:

- (1) Pick one or more user goals we hope our design addresses that are connected to one or more qualitative user requirements (goals and requirements explained in methods subsection above).
- (2) Recruit five participants from our target stakeholder group.
- (3) Use informed consent from each participant, letting them know they can stop at any time.
- (4) Ask each participant to use our design prototype to accomplish the goal while performing a think-aloud.
- (5) Wizard of oz the prototype based on the participant's actions.
- (6) Create individual interpretations of the participant's actions and utterances

Before starting the simplified user testing, we informed each participant of the purpose of our user testing, that is, evaluating how useful and usable our designs are. Following, we informed each participant of the think-aloud style of the simplified user testing, namely that they must keep speaking their thoughts out loud throughout the whole interview. Moreover, we informed them that they cannot ask us for questions since we are not able to intervene, and that they should simply keep interacting with our prototype until they complete their assigned goal. The specifics of each individual simplified user testing slightly differed, some being held online and others in person. For 2 of 5 testings held online, we additionally obtained consent to record each participant so that we can later take notes and create individual interpretations of their actions and utterances. On average, our simplified user testings lasted around 32 minutes.

To ensure quality control of tasks, we all agreed beforehand to use the exact same goal for each participant: Obtain a short bullet-point summary of a research paper to be used for an upcoming presentation. Additionally, we all utilized the exact same prototype, ensuring that all participant would have access to the exact same features. Finally, we also agreed on how to wizard of oz the prototype, ensuring that within each simplified user testing, our prototype would behave identically given the same instructions.

7.2.3 Participants. For simplified user testing, we decided to recruit 5 participants, the typical upper bound that is recommended for such testing. Each group member recruited one participant such that they are fully relevant to our focus group: multilingual researchers that utilize LLMs in their work. No additional incentives were given to the participants.

ID	Field	Value
P1	Role / background	Financial law professor
	Age	50
	Gender	Female
	Mode (online / in-person)	online
	Location of testing	Participant's office
	Languages	Native Macedonian; advanced English
	LLM / task experience	Regular LLM use for academic/admin writing, brainstorming, clarifying concepts, translation, and literature review.
P2	Role / background	Software developer / researcher
	Age	24
	Gender	male
	Mode (online / in-person)	online
	Location of testing	India
	Languages	Native Hindi; fluent English (often uses Hinglish in ideation).
	LLM / task experience	Uses LLMs for brainstorming, writing, reviewing, and surveying research papers.
P3	Role / background	Statistics graduate student
	Age	23
	Gender	Female
	Mode (online / in-person)	in-person
	Location of testing	Leinweber
	Languages	Bilingual Chinese and English.
	LLM / task experience	Frequently uses generative AI for research, including paper reading and academic writing.
P4	Role / background	Predoctoral research fellow in business economics
	Age	25
	Gender	Female
	Mode (online / in-person)	in-person
	Location of testing	Leinweber
	Languages	Fluent English; more comfortable in Chinese.
	LLM / task experience	Uses ChatGPT and other AI tools for academic and research tasks; often searches academic topics in Chinese.
P5	Role / background	Medical graduate student and researcher
	Age	32
	Gender	male
	Mode (online / in-person)	in-person
	Location of testing	Leinweber usability test lab
	Languages	Native Burmese; English as secondary academic language.
	LLM / task experience	Advanced experience with medical literature; regularly reads English journals and was chosen to test TEA's accessibility and translation support.

Table 6. Participant information for User Testing

Within each simplified user testing, we obtained informed consent from each participant, informing them that they can stop the testing at any point. Moreover, for P1 and P2, we additionally obtained consent to record them and their screen so that we can take notes of the testing afterwards.

7.2.4 Results. In the simplified user testing phase, we invited participants from diverse language and research backgrounds (including Chinese, Hindi, Macedonian, etc.) to complete the same goal

using a TEA paper prototype. This goal yielded a short bullet-point summary of a research paper for an upcoming presentation. The entire test employed a think-aloud approach, with researchers simulating system responses in real-time using the Wizard of Oz mode to observe participants' actual thought processes and operational paths.

1. Overall Observations and Common Behavioral Patterns

Based on the records of all individual reports, participants' behavior generally exhibited the following characteristics:

- (1) **Initial Exploration and Uncertainty.** Almost all participants displayed exploratory behavior and brief hesitation when initially using the system. For example, Marko's participant repeatedly clicked the arrow button during the input phase and verbally expressed confusion: "*Ughh, where can I click... I'm clicking on the arrow.*" Another participant directly asked, "*Where can I give the paper?*" when faced with the interface. These observations indicate that users were unable to intuitively identify the input or upload entry points in the early stages of the task, suggesting that the system lacked sufficient **visibility and operational guidance**.
- (2) **Semantic Confusion Regarding Key Functions.** In multiple reports, users expressed misunderstanding of system terminology such as *Tone Match*, *Length*, and *Complexity*. One participant, upon seeing the interface, directly asked, "*Tone match, length, translate... what???*" Another Hindi user pointed out that the system-generated content "*this is not casual*," but could not find an entry point to modify the tone. This reflects a discrepancy between terminology design and users' mental models, leading users to re-enter prompts rather than utilize existing tools for correction.
- (3) **Misuse and Delayed Discovery of Core Interactive Functions.** Several participants initially misunderstood the purpose of the *Highlight* function. As one participant mentioned in Xinyu's report: "*I thought the highlight icon was just for marking text... Oh, it actually opens another window!*" Furthermore, some users were unable to determine the actual function of the slider, remaining uncertain even after repeated adjustments: "*I can't tell whether it really changed the style.*" This indicates unclear metaphors and a lack of immediate feedback in key interactions.
- (4) **Cognitive Burden During Language Switching.** Multilingual users frequently switched languages to confirm translation accuracy. For example, a participant in Xinyu's session repeatedly switched between the English and Chinese interfaces to verify the translation's accuracy. Another participant, after generating a Macedonian summary, remarked: "*These sentences are very poorly written... I'll rewrite them myself.*" This feedback reveals a significant gap in the system's cross-language consistency and translation quality, forcing users to exert additional cognitive effort to correct the output.

Taking into account all individual observations and verbal feedback, it can be concluded that the current design fails to adequately meet the following two core needs:

- **R4:** The results show that the system's output style is inconsistent across different languages and lacks a clear tone control feedback mechanism. For example, a Hindi user stated that the system failed to convey the desired "casual" tone, while a Macedonian participant directly rewrote the generated content.
- **R8:** Although the system provides sliders for adjusting complexity and tone, the lack of real-time feedback or register hints makes it difficult for users to determine whether the adjustments are effective. As one user commented after testing: "*It looks simpler now, but I'm not sure it's the tone I wanted.*"

Furthermore, multiple reports highlighted the lack of status indicators and undo functionality in the system, leaving users uncertain whether a task was being performed or how to revert to a previous version.

2. Subjective Reflection and Improvement Directions

Overall, simplifying user testing revealed shortcomings in the current TEA prototype regarding "language consistency," "operational transparency," "semantic clarity," and "user control." These issues reflect design gaps in the system's interpretability and usability. Based on the aforementioned problems, the following improvement suggestions are proposed:

Improvement Recommendation 1: Enhance Feedback Visibility and Terminology Comprehensibility. Add real-time status prompts (e.g., "*Generating summary...*", "*Simplifying output...*") during upload, slider adjustment, and translation processes, allowing users to understand the system's progress at any time. Provide concise tooltips or mini-examples for features such as *Tone Match*, *Complexity*, and *Length*—for example, "*Tone Match adjusts formality and phrasing based on academic tone.*"

Improvement Recommendation 2: Increase User Control and Freedom. Incorporate an undo or backward feature that allows users to revert changes or view history of generated texts, and make exit buttons more visible within pop-up menus. This will enhance control, reduce frustration, and improve the overall sense of system transparency.

8 FINAL DESIGN AND FUNCTIONAL HIGH-FIDELITY PROTOTYPE

Following our heuristic evaluation and simplified user testing phases, we developed and evaluated a high-fidelity functional prototype of Rosetta, our proposed system for supporting bilingual researchers in receiving generative AI assistance. The heuristic evaluation (Assignment 4) highlighted usability issues surrounding feature discoverability and clarity. Simplified user testing highlighted challenges in understanding of core functions, particularly around terminology like *Tone Match*. Participants expressed confusion about how to access key features. These findings directly informed our design decisions for the functional prototype. We implemented a highlight popup tooltip feature that surfaces three actions—Simplify, Translate, and Tone Match—specifically addressing the operational transparency and discoverability issues identified in prior evaluations. The tooltip appears automatically when users highlight portions of AI responses, employing an interaction pattern familiar from tools like Google Search AI Overview to reduce learning curves. Each action generates a new message displaying the transformed text alongside the original, eliminating the need for undo functionality while preserving user control and comparison capability. This design specifically addresses the original user requirements around efficient language switching (R11, R12) and tone adjustment that emerged from our contextual inquiry with bilingual researchers.

Upon implementing the high-fidelity prototype of Rosetta (after Rosetta Stone, of course), we conducted an evaluation detailed in this report to understand the extent to which it addresses identified user requirements related to helping bilingual users navigate core research tasks reflecting their multilingualness with more efficiency and ease.

Key Functionality of Rosetta

Rosetta offers a UI similar to common general-purpose generative AI tools used by the general public. Users may prompt via the conversational UI. We implemented the tooltip menu feature, offering actions tailored to the needs, preferences, and uses of bilingual researchers highlighted in contextual inquiry sessions.

Users may access this by highlighting parts of the response. The tooltip pops up automatically, allowing users to choose from one of the three actions offered.

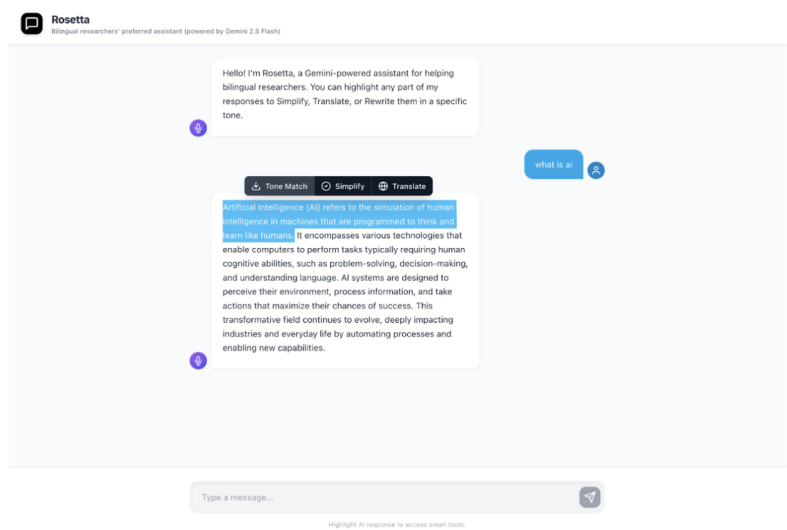


Fig. 18. Functional High-Fidelity Prototype of Rosetta with Highlight-Popup-Tooltip Implemented

Indeed, the decision to offer the "Tone Match", "Translate", and "Simplify" features in this floating tooltip was made following findings illuminated by our heuristic evaluation and simplified user testing where users struggled with finding and understanding this feature and actions it offers. The UI of our floating tooltip menu is unobtrusive and easily discoverable. The system provides reminders in many places (near chat bar, Rosetta's first message, etc) to remind users that they may highlight Rosetta's responses to access additional features. The tooltip is similar to many tooltips offered in various systems, including generative AI-enabled tools like Google Search AI Overview, for example.

This implemented interaction workflow allows for both immediate and multi-step AI transformations. When a user selects an action from the tooltip (such as Simplify, Tone Match, or Translate), the system automatically writes and executes a prompt (without the user having to engage in the chat bar) and sends it to Gemini API; the output is then rendered to the user as a new, distinct message, allowing users to compare the original output with the modified version and transform the original text, or this new transformed text snippet, as many times as they wish. This, again, addresses specific challenges raised in heuristic evaluation and simplified user testing dissected in the last assignment.

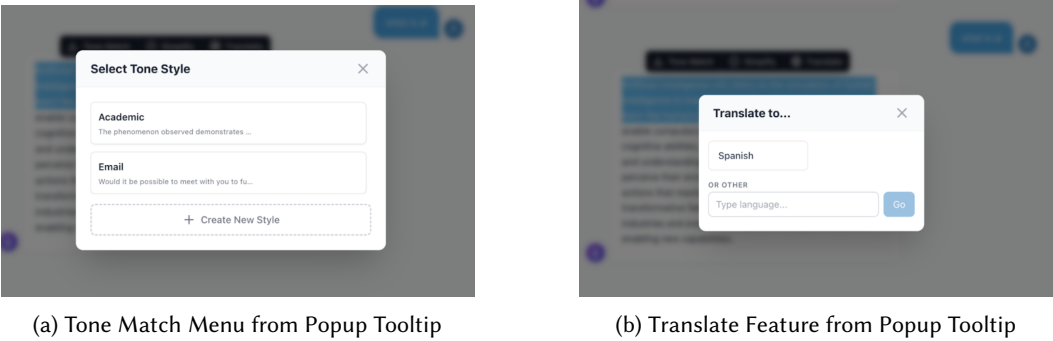


Fig. 19. Tone Match action (Figure 2a) allows users to match the tone of parts of AI response to a more appropriate tone (Academic and Email are available by default, addressing the particular use cases and needs of researchers; users may also create new styles (as shown in Figure 2b)

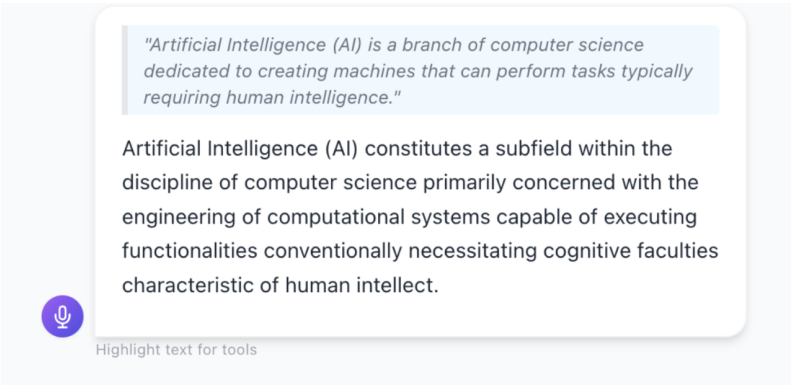


Fig. 20. Rosetta transforms and returns them as new “messages” that quote the user-highlighted snippet

9 USER EVALUATION

9.1 Purpose

This quantitative evaluation was conducted to assess the final functional high-fidelity prototype described in the previous section against user requirements devised according to grounded understanding of bilingual researchers in engaging with generative AI tools for assistance in research tasks.

Concretely, the evaluation compared an application offering the developed feature with one that, similar to existing general purpose generative AI tools, does not offer the implemented tooltip; they were compared according to two requirements to determine the usability and effectiveness in routine tasks (informed by contextual interviews) like adjusting tone for different research related tasks and translating particular snippets into languages. Indeed, the goal is that these implemented features address specific challenges observed during the prompting phase (where text entry for interaction with most generative AI tools take place).

9.2 Method

This study evaluates the against our updated requirements R11 and R12, both grounded in context of use. These requirements are:

- **Requirement 1.** Users should switch between different styles of writing for research more efficiently and with decreased cognitive demand.
- **Requirement 2.** Users should switch between languages when discussing research related topics more efficiently and with decreased cognitive demand.

To address these requirements, regarding efficiency and cognitive demand of switching writing style necessitating different tones as well as switching language of use, we operationalize efficiency via task completion time and cognitive demand through task load self-reported using the NASA Task Load Index (NASA-TLX). Because we are interested in the difference in efficiency and cognitive demand of a user when using our implemented prototype versus baseline, we conduct a within-subject study where, for each task, participants experience both the baseline and prototype system (in counterbalanced order to prevent ordering effects). Each requirement (tone adjustment and language switching) will be addressed with a short task (detailed in the Tasks and Procedures section).

Concretely, this quantitative evaluation entails a within-subject user study evaluating the following dependent variables (DVs):

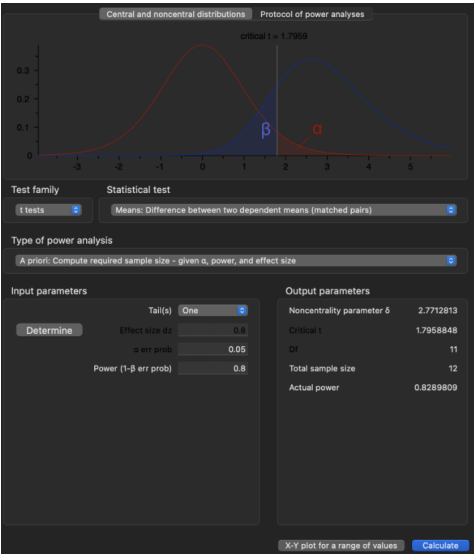
- **Dependent Variable 1:** Task completion time (minutes:seconds)
- **Dependent Variable 2:** Task load measured by taking the average of NASA Task Load Index (NASA-TLX) items reported on a 7-point scale

Here, the independent variable (IV) is the system the participant uses to carry out tasks. The treatment system offers the implemented highlight-popup-action menu; the baseline system is an identical application without that feature implemented; this minimizes the introduction of potential confounding variables by keeping all other aspects of the interface constant. Detailed in the Apparatus subsection, both resemble standard general purpose generative AI tools used by our target group for research tasks (ChatGPT, Claude, Gemini, etc).

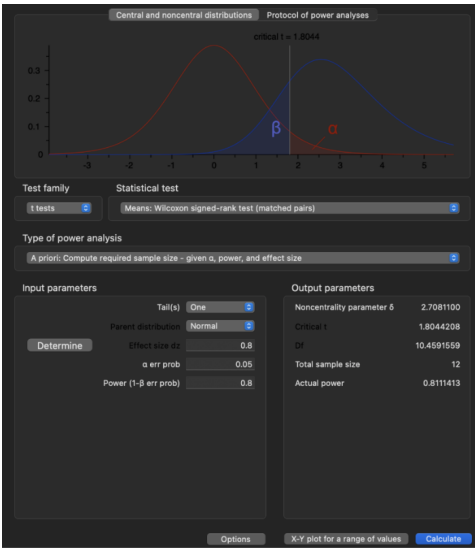
To examine differences in task completion time, we will use one-tailed paired-t-tests. We use paired-t-tests because the same participants complete tasks in both conditions, making the samples dependent. We assume the differences in task completion times will be approximately normally distributed given our sample size. We expect that our highlight-popup-action menu allows our prototype to result in reduced task completion time; as such, a one-tailed test is appropriate. We chose conventional values for alpha (0.05) and power (0.8) used by similar studies testing effectiveness of proposed text entry methods in HCI research. Our choice of effect size (large; Cohen's $d = 0.8$) reflects our confidence in our implemented feature that is informed by contexts of use and developed according to results of iterative evaluation, as well as constraints imposed by having a highly specific user group (bilingual researchers) that greatly constrains the recruitment pool for this evaluation study. Using G*Power, we determined that the required sample size with these parameters is $n=12$.

To evaluate differences in task load, we will use one-tailed Wilcoxon signed-rank tests. We use Wilcoxon signed-rank tests because we do not assume the NASA-TLX scores will follow a normal distribution; this non-parametric test is appropriate for ordinal scale data like Likert-type responses. Again, we expect that through offering actions tailored to specific challenges that cognitively bottleneck bilingual researchers, our prototype, when compared with the baseline, will result in reduced task load; this makes the one-tailed test appropriate. To examine these differences, we use alpha (0.05), power (0.8), and effect size (large; $abs(r) = 0.8$) chosen for the same reasons as

those outlined previously. Using G*Power, we determined that the required sample size with these parameters is $n=12$.



(a) One-tailed paired t-tests



(b) One-tailed Wilcoxon signed-rank tests

Fig. 21. G*Power Calculations

9.3 Apparatus

Technical Implementation: React Application with Gemini API

Rosetta is developed as a React application using TypeScript allowing users to receive assistance from Gemini 2.5 Flash. The system utilizes Gemini API and Google GenAI SDK to manage stateful chat sessions. The text selection feature is implemented via the ChatBubble component and the floating tooltip, spawned when text is highlighted, is implemented in the ‘TooltipMenu’ component. When a user clicks on an action available in the popup menu, the system constructs a prompt that combines the user’s highlighted text with the selected, pre-configured instruction (e.g., "Translate this text into Spanish") and sends it to the Gemini API via the ‘processTextWithTool’ service.

Implementation of the “Baseline” interface for evaluation study

In addition to developing the high-fidelity prototype offering the implemented tooltip menu, we also developed a “baseline” system that is identical to the former but without the tooltip menu available. When parts of Rosetta’s response are highlighted, the highlight popup menu does not appear. Contrast this with Figure 1, where the popup tooltip menu is highlighted when the user highlights part of Rosetta’s response. To avoid possible human errors during the experiment, we implemented these as separate applications available on different links.

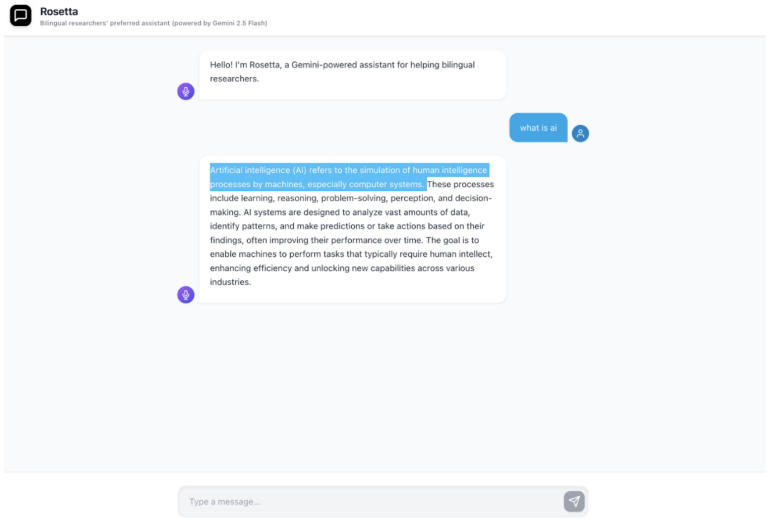


Fig. 22. Baseline Interface (no highlight popup menu)

9.4 Tasks and Procedures

This study evaluated the efficiency and cognitive load of the implemented popup menu feature through two tasks; for each, we had participants perform two conditions in counterbalanced order: the baseline system (without popup menu) and the treatment system (with popup menu). Recall that the evaluation study aims to evaluate the extent to which the popup menu feature reduces cognitive load and improves task completion time when users need to adjust writing tone or translate content during research activities.

Task 1: Tone Adjustment

Objective: The objective of this task is to evaluate the extent to which the popup menu feature reduces cognitive demand and improves efficiency when users need to adjust writing style under different academic contexts.

Requirements Evaluated: Requirement 1

Task: Participants are told that they just came up with an idea they want to do for their lab group while watching videos about agentic scientific discovery. They are interested in whether making a similar agentic tool can help people in their research lab with reading and synthesizing papers more effectively. They are asked to enter the following prompt into the system (instructing the LLM to edit a message they will send to their PI): “Edit msg i want make agentic tool make auto lit review for wkly reading group” Since the prompt is quite generic, the AI response will likely still not be in an appropriate tone. As such, participants, upon receiving the AI response, will then be instructed to modify this message so it is suitable to send to a professor (i.e., adjust the tone to be more formal and academically appropriate). Task completion time will be measured from when the instruction is given until the participant indicates they have successfully completed the tone adjustment. The facilitator begins the timer when participants receive the instruction and stops when they verbally confirm completion and are satisfied with the AI’s revised message.

Task 2: Translation (Testing Requirement 2)

Objective: The objective of this task is to evaluate whether the popup menu feature reduces cognitive load and improves efficiency when users need to translate research content between languages.

Requirements Evaluated: Requirement 2

Task: Participants will be asked to enter the following prompt into the system: “mitigating LLM hallucinations” Participants will then be instructed to translate the first sentence of the AI’s response into Spanish; they simulate a scenario where they are an international student from Spain and are interested in sharing the information given in the first sentence of the AI response to a friend back home who is not fluent enough in English to understand this technical content. Task completion time will be measured from when the translation instruction is given until the participant confirms, verbally, that they have completed this translation task. Again, the timer begins when participants receive the instruction and stops when the translated content is displayed.

Evaluation Procedure

- (1) **Introduction and Informed Consent.** Participants are briefed on the purpose of the study and reminded that the task evaluates the system rather than their personal ability. Informed consent is obtained verbally and recorded through a Google Form. Participants are also informed that they may withdraw at any time.
- (2) **Participant Survey.** Before beginning the tasks, participants complete a brief demographic survey.
- (3) **System Familiarization.** Participants receive a 2–3 minute walkthrough of their first assigned system (baseline or treatment). The walkthrough includes an example of the facilitator entering text prompts (e.g., “what is AI”). For systems with the popup feature, the facilitator demonstrates how to highlight AI responses and use the popup menu. They show what “simplify” (simplifying language), “translate” (translating the first sentence into Spanish), and “tone match” (adjusting to formal academic tone) do.
- (4) **Tone Adjustment Tasks.** Participants complete both tone adjustment tasks in sequence. To minimize carryover effects, a counterbalanced design is used: half of the participants begin with the baseline system, and the other half begin with the treatment system. Within each condition, participants receive instructions, are encouraged to think aloud, are timed with a stopwatch (minutes:seconds), and complete a NASA-TLX survey via Google Form immediately after each task.
- (5) **Translation Tasks.** Participants perform both translation tasks in counterbalanced order. Within each condition, they receive instructions and context, complete the task while being timed, and fill out the NASA-TLX survey immediately afterward. Participants are encouraged to talk aloud as they work.
- (6) **Post-Task Feedback.** After completing all four tasks, participants are thanked for their time and asked to provide qualitative feedback about their experience with the tasks.

NASA-TLX Questionnaire

Immediately after each individual task, participants will complete the NASA-TLX questionnaire to assess task load. We will average the responses to these five items to measure task load. This results in four NASA-TLX measurements per participant (two tasks × two conditions). The following items from the NASA-TLX Questionnaire were used:

- **Mental Demand:** How mentally demanding was the task? (1 = very low; 7 = very high)
- **Temporal Demand:** How hurried or rushed was the pace of the task? (1 = very low; 7 = very high)
- **Performance Demand (reverse scored):** How successful were you in accomplishing what you were asked to do? (1 = failure; 7 = perfect)
- **Effort:** How hard did you have to work to accomplish your level of performance? (1 = very low; 7 = very high)

- **Frustration:** How insecure, discouraged, irritated, stressed, or annoyed were you? (1 = very low; 7 = very high)

Data Quality Assurance

To ensure data quality, task instructions are delivered using a standardized script and used a spreadsheet to coordinate conditions for counterbalancing. Measurements were synchronized on a shared spreadsheet.

9.5 Participants

With n=12 needed for sample size and five team members, each team member was responsible for two to three participants. Each participant did two tasks and experienced both baseline and treatment in each. In each task condition, data was collected via a 7-point NASA-TLX survey (with physical demand excluded). The responses were averaged, with the performance being reversed scored. Additionally, trial time is reported in minutes (m:s format).

Inclusion and exclusion Criteria

Our project seeks to address the particular challenges encountered by bilingual researchers when conducting research. Thus, we sought participants who conduct research for professional purposes (e.g., graduate students are considered as research professionals) or are conducting research for professional training en route to pursuing a profession in research (e.g., masters students conducting research on a regular basis interested in pursuing PhD programs). Additionally, we sought participants who deem themselves to be bilingual: in American universities, these tend to be individuals who have native languages other than English that they may have more comfort in.

Table 7. Participants Information for Quantitative Testing

Participant	Fluent language	Main academic research language	LLM usage frequency
P1	Chinese and English	Chinese	Weekly
P2	Chinese and English	Chinese	Weekly
P3	English and Mandarin and Cantonese	English, Mandarin	Daily
P4	Hungarian	English	Weekly
P5	English and Hindi	English	Weekly
P6	Hindi and English	English	Daily
P7	Mandarin	English (American)	Daily
P8	Chinese	English	Rarely
P9	Hindi and English	English for professional communication	Daily
P10	English and Serbian; Croatian and Macedonian	English	Daily
P11	Macedonian and English	English	Daily
P12	Persian and English	English	Daily
P13	Urdu and Punjabi	English	Daily
P14	Mandarin, English	English	Weekly

9.6 Results

We used the Rosetta high-fidelity feature prototype for quantitative evaluation, which addresses the efficiency and cognitive load challenges faced by bilingual researchers in intonation and translation tasks through a highlight-pop-action menu feature. Based on the raw data on task completion times from 15 participants, the highlight-pop-up-action menu (Treatment) consistently demonstrated an advantage in improving the efficiency of users performing intonation adjustment and translation tasks. Without this feature (Baseline, Task 1, and Task 3), participants generally spent more time, with significant individual differences in completion times; for example, the longest completion time for the intonation adjustment task reached 5 minutes and 30 seconds. In contrast, after using the prototype system (Treatment, Task 2, and Task 4), the vast majority of participants showed a trend towards shorter completion times. Particularly in the translation task (Task 4), many participants significantly reduced their completion time compared to the baseline conditions (Task 3), fully demonstrating the potential effectiveness of the pop-up menu feature in simplifying operational steps and accelerating the completion of specific research-related tasks.

Table 8. Participant completion time (minutes:seconds) for different tasks

Participant ID	Task1 (baseline-Tone)	Task2 (Treatment-Tone)	Task3 (baseline-translat)	Task4 (Treatment-transl)
1	3:01	2:09	1:21	0:39
2	5:00	4:23	2:29	2:32
3	5:27	2:03	1:29	0:58
4	2:50	3:55	3:16	2:51
5	5:14	4:50	6:13	5:27
6	3:44	4:13	2:57	3:19
7	5:30	3:40	3:10	1:25
8	5:22	4:23	3:50	2:10
9	4:20	3:47	5:12	1:26
10	3:26	2:51	0:41	0:48
11	4:29	4:01	0:55	0:34
12	1:46	2:34	0:30	0:22
13	2:55	2:49	1:35	1:55
14	2:38	1:17	1:01	1:11

Table 9 presents all inferential statistical test results for both tasks.

For the tone adjustment task, both objective performance metrics and subjective workload measures converged to show clear benefits of the pop-up menu system. As shown in Figure 23 (left), participants completed the task significantly faster when using the prototype system ($M = 201.07s$) compared to the baseline ($M = 238.71s$). The consistent height difference between the two bars and their non-overlapping error ranges visually align with the statistical result, indicating a meaningful reduction in task completion time. The paired t-test confirmed this improvement ($p = 0.0284$), with a medium effect size (Cohen’s $d = 0.56$), suggesting that the feature produced stable and practically relevant efficiency gains across participants.

Table 9. Summary of Statistical Test Results for Both Tasks

Task	Measure	Test	Statistic	p-value	Effect Size
Tone Adjustment	Time	Paired t-test	$t = 2.09$	0.0284	$d = 0.56$
Tone Adjustment	NASA-TLX	Wilcoxon	$W = 81$	0.0368	$r = 0.48$
Translation	Time	Paired t-test	$t = 2.17$	0.0247	$d = 0.58$
Translation	NASA-TLX	Wilcoxon	$W = 59.5$	0.0538	$r = 0.43$

This improvement is further supported by the distribution plot in Figure 24 (left panel), which shows that most participants had positive time savings (baseline minus treatment), with a mean of 37.64 seconds. Although some participants demonstrated smaller or even slightly negative differences, the majority clustered around a positive improvement zone, reinforcing the robustness of this effect.

In terms of cognitive load, NASA-TLX scores (Figure 23 , right) also showed a statistically significant reduction when participants used the prototype ($p = 0.0368$). The bar graph reveals a clear downward shift from a mean of 3.20 (baseline) to 2.49 (treatment). The Wilcoxon signed-rank test and the medium effect size ($r = 0.48$) indicate that the prototype effectively reduced the perceived mental demand, effort, frustration, and temporal pressure of participants. The distribution of TLX score changes in Figure 24 (right panel) also shows that most participants report fewer points of workload under the prototype condition. Together, these findings provide strong evidence that the pop-up menu feature meaningfully supports users in tone adjustment tasks to improve both efficiency and cognitive burden.

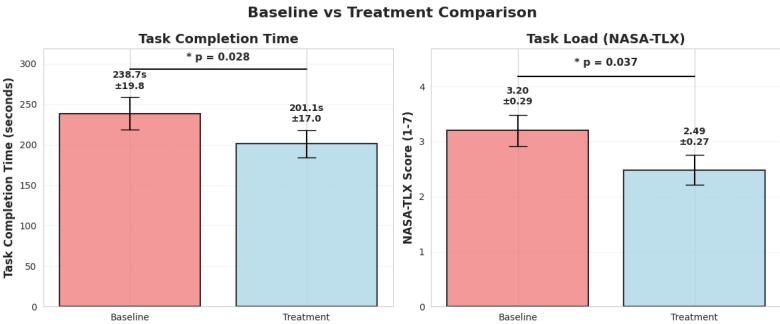


Fig. 23. Quantitative Evaluation Results - Tone Task

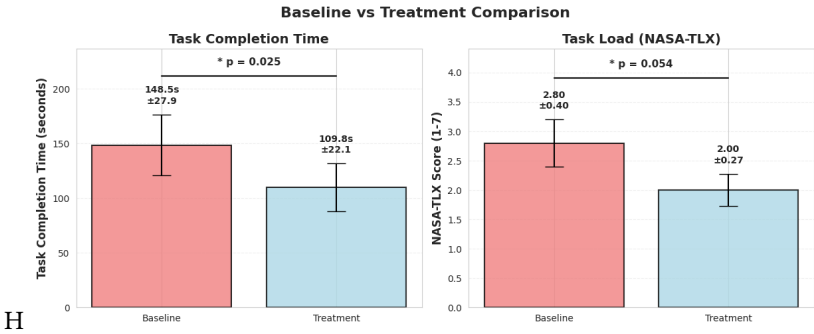


Fig. 25. Quantitative Evaluation Results - Translation Task

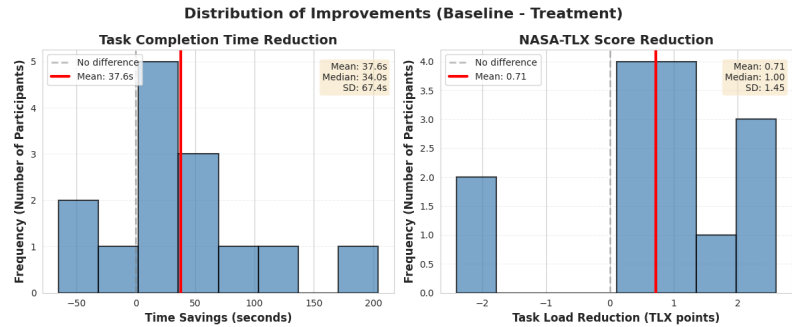


Fig. 24. Quantitative Evaluation Results - Tone Task

For the translation task, task completion time again showed a clear and statistically significant improvement. As illustrated in Figure 25 (left), the treatment condition yielded substantially faster performance ($M = 109.79s$) compared to the baseline ($M = 148.50s$). The paired t-test confirmed this reduction as significant ($p = 0.0247$), with a Cohen's d of 0.58, again falling within the medium effect size range. The distribution visualization in Figure 26 (left panel) similarly shows that most participants saved time under the treatment condition.

However, unlike the tone adjustment task, the NASA-TLX results for translation did not reach statistical significance ($p = 0.0538$), despite showing a numerically meaningful decrease (from 2.80 to 2.00). As depicted in Figure 25 (right), the bars do show a downward trend, but the variability across participants was greater, leading to overlapping error bars and reduced statistical certainty. This pattern is also reflected in the distribution plot in Figure 26 (right panel), where several participants reported equal or slightly higher cognitive load under the treatment condition. Together, these observations suggest that while the translation feature improved efficiency, users' subjective experience of workload may be less sensitive to interface support in this task, possibly because translation is already a familiar and low-ambiguity operation for multilingual researchers.

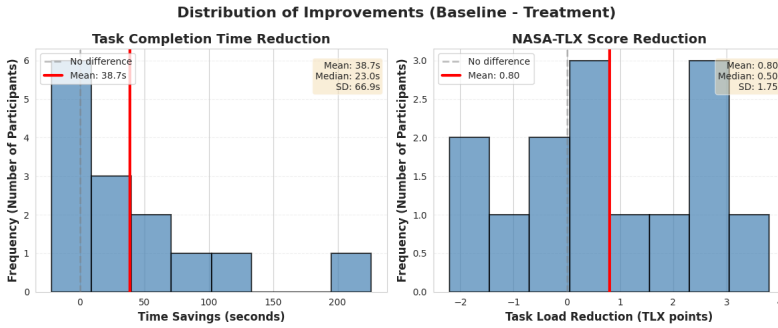


Fig. 26. Quantitative Evaluation Results - Translation Task

10 DISCUSSION

10.1 Initial Survey for Establishing Focus

10.1.1 *Interpretation of Survey Results.* Overall results from the survey suggest that, indeed, multi-lingual researchers make use of generative AI-enabled tools for research related tasks. Respondents reported using generative AI-enabled tools for a myriad of tasks, with the three most frequently reported ones being literature review, academic writing, and clarifying concepts. It is worth noting that all of these are quite cognitively demanding tasks that do demand a nontrivial amount of text entry. Notably, these are all related to engaging with different pieces of academic writing, whether consuming, producing, or deciphering. Regarding specific activities through which users perform text entry, the highest number of participants indicate that they structure their input via short prompts, details and copy pasting.

Survey results surface potential issues that users may experience when performing text entry in different languages to use generative AI capabilities in assisting them with research related tasks. First, when asked about the number of prompts required to get a satisfactory response in their native language compared with english, the highest number of participants (10/24) responded “much less”; this suggests potential reasons that may explain why a user may not want to prompt in English despite papers often being in English. However, when asked about what challenges users face in prompting in their native language, the highest number of respondents (11/24) reported “lower-quality or less relevant responses when using my native language.” This suggests that some multilingual users, perhaps, may have a mental model of generative AI models as being less capable in their native language that they, themselves, may be more comfortable using compared with English. To handle challenges that occur when prompting using their native language, the highest number of participants reported resorting to “rewrit[ing] or simplif[ing] prompts multiple times.”

Finally, survey results suggest potential directions for addressing these issues. Namely, when asked about what changes in interfaces, through which they perform text entry to interact with generative AI capabilities, could address potential communication challenges they encounter, the highest number of participants (12/24) selected “better multilingual support and translation within the interface.”

10.1.2 *Implications of results on scoping of the focus.* Taken together, initial survey results highlighted the potential directions this project pursued to better understand and characterize issued encountered by multilingual users when performing text entry to receive generative AI-enabled assistance for research-related tasks.

First, the study suggested that users do, indeed, type in multiple languages or have to make decisions involving tradeoffs between what languages they use when performing text entry, which they do through both short-formed and detailed prompting. Immediately, this led us to a more thorough examination of how users make these decisions—for instance, factors, considerations, or past experiences that may influence these decisions.

Second, study results offered insights on tactics multilingual users may engage in when faced with issues during text entry to interact with generative-AI enabled tools during research. However, due to fundamental limitations of surveys, current results do not sufficiently represent the distribution of tactics taken by users. For instance, consider how users may not remember what exact tactic they used for an issue they encountered a few months ago; contextual inquiry, or other forms of observation where participants will encounter challenges in real-time, may help provide more informative insights on tactics users actually engage in. Understanding tactics may help pinpoint what issues are so paramount that users feel the need to put in extraneous effort in resolving, rather than ignoring. Designs can then address these specific challenges, perhaps taking inspiration from recurring tactics users engage in.

10.2 Contextual Interviews for Understanding Context of Use

10.2.1 Interpretation of Consolidated Findings from Contextual Interview and New Insights Generated. Informed by insights from initial surveys, we conducted a series of contextual interviews to ground our understanding in the context of use. Collecting, interpreting, and consolidating these findings with our prior understanding allows us to work towards specifying appropriate user requirements. Indeed, our contextual inquiry with five multilingual researchers provided us with five grounded accounts of how users navigate language choices and experience breakdowns when prompting LLMs for academic research tasks. These findings build upon the initial survey by moving from self-reported patterns to observed behaviors in real research contexts.

First, new data collected in contextual interviews provide us with prospective helpful to understand processes underlying insights generated in the initial survey, that users rarely prompt in their native language. Specifically, results from the contextual inquiry illuminated the reasoning underlying these choices. We found that participants we observed do have default to English; rather, they deliberated according to specific task demands, perceived model capabilities, and cognitive comfort. Consider how P1 (a researcher with intermediate English proficiency) chooses to scaffold complex ideas in Chinese to clarify her thinking and would translate critical technical terms into English before prompting. P3, who is bilingual in Hindi and English, described thinking naturally in a combination of the two languages but typing exclusively in English due to script-switching friction and his belief that ChatGPT interprets English more reliably. Although consolidated interpretations from these sessions are not sufficient to construct comprehensive mental models of how these users think about language, they provide us with intuition helpful to consider moving forward.

Second, contextual interviews also shed light on major breakdowns that occur. Specifically, participants seem to encounter two dominant types of breakdowns: inaccurate translations that disrupt comprehension and mismatches between requested and delivered explanation depth. We can think of these breakdowns as miscommunications that occur due to the use of more than one language. Regarding the case of inaccurate translations, consider how P4 frequently received Bulgarian or Serbian translations instead of Macedonian, forcing manual verification or re-prompting in English. P2 observed that Korean summaries often exhibited overly formal, artificial phrasing that felt disconnected and, at best, quite strange. Indeed, it seems to be the case that these issues result in undermined trust and verification loops that result in heightened cognitive load and time expenditure. Similar issues emerge even when translations are technically correct, but the depth is a bit off.

For instance, consider how P1 requested concise summaries but received verbose, over-explained outputs. Current limitations of generative AI models when it comes to maintaining contextual awareness of user expertise, task stage, and disciplinary norms may explain this phenomena.

But how do bilingual users navigate these challenges? In our contextual interviews, we observe that participants seemed to have adapted to these recurring challenges. They have diverse tactics to mitigate breakdowns. P3, for instance, manually cross-checks AI-generated definitions against research papers (due to doubts about ChatGPT's accuracy for academic terminology). Indeed, it seems like these workarounds suggest needs for greater transparency and control over model behavior.

In addition to using specific tactics routinely, we considered what role their native language, as a tool, serves these users. We found participants describing their native languages as a form of cognitive anchor that affords them more natural and familiar thinking, understanding verification, and developing confidence before translating ideas to English. P1, for example, articulated how forming ideas in Chinese felt “more comfortable.” In doing so, she was able to clarify thoughts before struggling with English phrasing.

Ultimately, our contextual interviews also highlight the importance of attending to task characteristics where interaction resides. Early in research projects, participants used ChatGPT exploratively to build foundational understanding, tolerating less precision. Later, when refining drafts or preparing for supervisor meetings, they demanded higher accuracy and tighter alignment with disciplinary norms.

10.2.2 Implications of results on scoping of the focus. Together, the contextual interview results have updated our understanding, informed previously via our initial survey, of how bilingual researchers experience text entry when interacting with generative AI tools for research assistance. To recap, our consolidated findings hint at the underlying and typically unobservable thought processes underlying deliberation before entering text to interact with generative AI tools for research assistance, frustrating breakdowns that may be a consequence of miscommunication and may necessitate further prompting (and thus more text entry), existing workarounds that may provide intuition for expectations and standards, what text entry in English vs their native language entails and how they provide different purposes, and task-specific contextual considerations. These consolidated insights all ground our thinking in what the text entry experience is like for bilingual researchers using generative AI assistance for research.

Contextual inquiry findings provide intuition for specific critical aspects of the text entry experience to focus on. Specifically, we now have grounded accounts explaining the antecedents (cognitive deliberation about language choice, mental models of model capabilities, task-specific considerations) and consequences (mistranslations requiring verification loops, explanation depth mismatches necessitating repeated text entry, breakdowns in trust leading to abandoned interactions) of multilingual text entry to with generative AI tools for research support. With awareness of surfaced antecedents and consequences of bilingual users' text entry experiences, we are now able to identify missed opportunities in operationalizing users' deliberations (i.e., what language, how to prompt, why, etc) and other antecedents in mitigating existing limitations suggested by unideal consequences (e.g., breakdowns and ‘hacks’ to get around them).

10.3 Initial Design

We conducted a design critique process to understand potential value towards meeting user requirement of interactions proposed by team members in the context of affording better means for text entry in the context of supporting multilingual researchers in Anglophone research. First, we used the “I like, I wish, What if” framework to systematically evaluate five individual designs against user

requirements. We consolidated the reviews (detailed above) to ultimately arrive at a consolidated design that addresses pre-text entry and post-text entry interaction phases in prompting. Our team members arrived at recurring critique and ideas. First, we seem to find adaptive controls (particularly sliders for adjusting complexity, length, and tone) as a way to reduce cognitive load during text entry. Simultaneously, though, some critiques raised questions about overwhelming users with too many customization options. Second, voice input was found to be another viable mechanism for enabling natural multilingual expression. Third, members also considered highlighting-based interactions to provide strong support for their intuitive nature that reduces what people have to type in text entry. When consolidating, we arrived at the following ways of addressing the antecedents and consequences of text entry for bilingual researchers in interacting with generative AI tools. For antecedents, we will be addressing by using Ishika's voice input concept. For consequences, we will address via Marko's post-generation refinement tools; we significantly revised the tone match feature to accept user-provided writing samples rather than abstract tone descriptors, directly responding to reviewer confusion about how tone matching would work in practice. The consolidated design targets twelve of nineteen user requirements. It is worth noting that we particularly emphasized on those most directly tied to observed breakdowns during contextual inquiry. The pre-prompt voice interface addresses R3 (language switching without context loss), R11 (minimizing cognitive fatigue), and R12 (expressing ideas in clearer language). The post-prompt refinement tools address R4 (consistent tone across languages), R7 (quick clarification), and R15 (correcting misinterpretations) by providing granular control over AI-generated text without requiring complete re-prompting (painful, annoying, inefficient text entry). Our decision to make refinements occur directly on existing text, rather than generating new prompts, directly responds to the verification loops and re-entry cycles observed during contextual inquiry. By enabling in-place modifications through highlighting and sliders, the design reduces the cognitive burden of managing multiple response versions while maintaining conversational continuity.

10.4 Initial Design Evaluation and Improvements

The consolidated design decisions discussed in the previous section formed the foundation of the prototype we later evaluated with multilingual researchers. Our design critique process helped identify feature directions and anticipate several challenges that surfaced again during user testing, including uncertainty around tone-related controls, the need for clearer affordances for highlighting, and the demand for low-effort ways to refine AI-generated text. The prototype implemented these ideas through voice-based pre-prompt input, structured prompt generation, and post-generation refinement tools. However, evaluation showed that several aspects of the interaction still required improvement, particularly around terminology clarity, system transparency, feedback visibility, user control, and onboarding. These insights directly informed the next phase of design refinements.

User testing of the multilingual prompt-editing prototype revealed clear strengths and key usability issues. The prototype enabled non-native English-speaking researchers to voice ideas in their native language, review them as structured English prompts, and refine tone, complexity, and clarity.

Several usability gaps also emerged. Participants found interface labels such as Tone Match, Length, and Complexity unclear, suggesting a mismatch with their expectations. Many users overlooked interactive elements like the Highlight feature due to weak affordances and limited feedback. Multilingual participants additionally experienced cognitive load when switching languages, repeatedly verifying translations to ensure meaning was preserved.

10.5 Functional Prototype Evaluation

10.5.1 Addressing Initial Evaluation Findings to Implement High-Fidelity Prototype. Our evaluation of the initial design highlighted specific usability barriers that hindered the system's ability to provide improved means for effective text entry for multilingual researchers in prompting generative AI tools for assistance. Recall that findings highlighted ambiguity surrounding abstract tone descriptors and the low discoverability of highlighting interactions. These findings directly informed design decisions made in constructing our high-fidelity prototype. To address the confusion regarding tone controls, we shifted from abstract adjective selection to a concrete example-based input system, allowing users to anchor the AI's output to specific disciplinary writing styles such as "Academic" or "Email" formats, with the ability to create custom tone profiles. Moreover, we addressed the visibility issues associated with the highlight-to-edit interaction by implementing a pop-up menu that becomes immediately salient (high-contrast UI) upon text selection, accompanied by persistent interface reminders about the highlighting functionality. Indeed, our design choices were informed by prior insights aimed at reducing the ambiguity and mechanical friction of specific interactions taken by multilingual researchers in complex research tasks. Ultimately, this resulted in the implementation of a highlight-popup-tooltip feature that we offered in an interface that otherwise provides similar experience and functionality as popular general purpose generative AI tools such as ChatGPT, Claude, and Gemini that are currently in widespread use among our target population.

10.5.2 Assessment of User Requirements. To evaluate the efficacy of these high-fidelity implementations—the extent to which the features we implemented address the challenges surfaced in contextual inquiry—we focused on assessing two critical user requirements derived from our earlier contextual inquiry with bilingual researchers. Specifically, we evaluated whether users could switch between different styles of writing for research more efficiently and with decreased cognitive demand (Requirement 11), and whether users could switch between languages when discussing research-related topics more efficiently and with decreased cognitive demand (Requirement 12). These requirements emerged directly from observed breakdowns where participants struggled with tone mismatches and engaged in effortful manual translation processes. We conducted a within-subject controlled experiment with 14 multilingual researchers, comparing our prototype against a baseline interface without the highlight-popup-tooltip feature. Participants completed two tasks—changing the style of a casual research message to make it appropriate for a professor and translating technical content to Spanish—while we measured task completion time in seconds and perceived cognitive load via averaged NASA-TLX scale items. Statistical significance was determined using one-tailed paired t-tests for temporal data and one-tailed Wilcoxon signed-rank tests for ordinal workload data, with counterbalancing employed to control for order effects.

10.5.3 Interpretation of Quantitative Findings. The evaluation of the tone modification requirement yielded robust evidence supporting the efficacy of our design interventions. Participants demonstrated a statistically significant reduction in task completion time ($p = 0.0284$, Cohen's $d = 0.558$) when using the highlight-popup menu compared to the baseline, completing tasks an average of 37.64 seconds faster. Furthermore, the self-reported task load was significantly lower ($W = 81.0$, $p = 0.0368$, $r = 0.478$). These results suggest that enabling in-context refinement, specifically the ability to highlight and adjust tone without the need to formulate entirely new text prompts, successfully offloads the cognitive burden associated with stylistic code-switching between informal ideation and formal academic communication. By allowing users to manipulate existing text rather than generating new inputs from scratch, the interface seemingly aligns more closely with the iterative nature of academic writing and eliminates the prompt reformulation step

that contextual inquiry reveals as a significant source of friction and burden that sometimes leads to abandonment entirely. Regarding the translation requirement, results indicated a similar gain in temporal efficiency, with participants completing translation tasks significantly faster using the prototype ($p = 0.0247$, Cohen's $d = 0.579$), representing a mean reduction of 38.71 seconds. However, while there was a decreasing trend in perceived task load, it did not reach statistical significance ($p = 0.0538$). A potential explanation for this phenomenon is that while our interface accelerated the mechanical act of obtaining a translation, it could not eliminate the inherent cognitive demands of semantic verification that multilingual researchers engage in to ensure accuracy. Multilingual researchers often feel compelled to mentally cross-reference AI-generated translations against their own linguistic knowledge to ensure fidelity of meaning, a cognitive process that remains demanding regardless of how quickly the interface delivers the text. Synthesizing these findings offers new grounded insights into the experience and challenges multilingual researchers face when performing text entry for generative AI assistance. Specifically, the quantitative evaluation suggests that mechanical friction, operationalized as time, is more easily mitigated by interface design than certain forms of cognitive friction (measured through task load index), particularly verification load pertaining to translation. Ultimately, the significant reduction in time across both tasks validates the highlight-popup-tooltip interaction technique as an improved means for conversational re-prompting among multilingual researchers, providing empirical evidence that eliminating prompt reformulation cycles addresses genuine bottlenecks in existing generative AI interfaces. However, we note the limitations of our evaluation: short-term testing may not fully capture the cognitive demand of real-world tasks or reveal longer-term effects such as learning curves or integration into authentic extended research workflows. Future iterations must consider a broader view of the research workflow multilingual researchers take, for example; this may entail switching between different applications like Google Chrome, Zotero, word processors, etc. Indeed, future iterative refinements in design and future evaluation must consider how efficiency gains and cognitive load reductions most likely vary across different research activities that bilingual users engage in when prompting generative AI tools.

11 CONCLUSION AND FUTURE WORK

11.1 Initial Survey for Establishing Focus

11.1.1 Brief summary of outcomes and new knowledge derived. Our initial study revealed that multilingual users, indeed, encounter issues stemming from text entry when interacting with generative AI tools for research-related tasks. Results suggest that these users often face challenges regardless of what language (their native language, English, or some combination) they prompt in. As such, they have to make tradeoffs, address challenges, or sacrifice in receiving suboptimal assistance from generative AI tools. This new knowledge allowed us to significantly narrow our scope, as discussed in previous sections.

11.2 Contextual Interviews for Understanding Context of Use

11.2.1 Brief summary of outcomes and new knowledge derived. Our contextual inquiry study provided a grounded understanding of how multilingual researchers reasoned about, engaged in, and developed expectations for interactive interactions with generative AI tools, mediated through text entry for research assistance. Results indicated the factors they considered when deliberating what language to enter. These were not fixed preferences; instead, they were decisions shaped by cognitive comfort, perceived model reliability, and task demands. Grounded interpretations also revealed existing breakdowns that stemmed from and contributed to iterative text entry, as well as how participants reacted to them. Workarounds such as code-switching, verification loops, and

manual rephrasing were often employed when breakdowns like mistranslations of non-negotiable terminology or mismatches in explanation depth occurred.

By integrating interpretations distilled from these grounded understandings with insights generated through the initial survey analysis, we surfaced the antecedents and consequences of multilingual text entry experiences. This integrated, and thus abstracted, view illuminated where current text entry interfaces missed opportunities to operationalize users' deliberations and mitigate existing limitations. This narrowed focus, in turn, allowed us to identify specific design targets within the text entry experience.

11.3 Initial Design

Our design critique and consolidation process revealed that multilingual text entry challenges could be addressed by targeting the aforementioned antecedents through pre-prompt voice input and the consequences through post-prompt refinement tools. The critique process we conducted surfaced new knowledge about the design trade-offs involved. It appeared that while a comprehensive approach to feature coverage could theoretically address more user requirements, more principled and focused designs targeting specific interaction moments proved more usable, less cognitively overwhelming, and better suited for supporting bilingual researchers encountering issues during text entry with generative AI for research assistance. Voice input effectively addressed antecedents of text entry challenges (language deliberation, cognitive fatigue), while highlighting-based refinement tools addressed consequences (verification loops, re-prompting cycles). Together, the consolidated design targets covered twelve of nineteen user requirements, prioritizing those most directly tied to observed breakdowns from contextual inquiry. This selective approach, informed by critique feedback, acknowledged that not all requirements could or should be addressed in a single interface.

Building on the consolidated low-fidelity prototype, we conducted usability evaluation to assess the extent to which our design decisions operationalized the deliberative processes and mitigated the breakdowns observed during contextual inquiry. Evaluation observed whether the designs offered reductions in verification loops, re-entry cycles, and cognitive overhead.

11.3.1 Design Evaluation and Improvements. This evaluation underscored both the promise and the limitations of our initial prototype in supporting multilingual researchers as they prompt and refine AI-generated academic text. The prototype demonstrated value by enabling users to speak ideas in their native language, generate structured English prompts, and refine tone and clarity through targeted adjustments. However, user testing also revealed several areas that require further attention, including unclear terminology, limited system feedback, weak affordances for interactive elements, and cognitive effort associated with switching between languages and verifying translations.

Moving forward, we integrated the design improvements identified through our evaluation into the next iteration of the prototype. This included developing clearer onboarding and support materials, adopting more intuitive labels and terminology, adding real-time feedback prompts to increase system transparency, strengthening user control through mechanisms such as undo/redo and edit previewing, and enhancing the discoverability of features like highlighting and refinement adjustments. These refinements were incorporated into the final functional prototype to create a more guided, transparent, and supportive interaction experience for multilingual users.

11.4 Functional Prototype Evaluation

The quantitative evaluation of the functional prototype provided empirical evidence that the implemented highlight–tooltip–popup feature did, in fact, reduce certain forms of cognitive and mechanical friction associated with re-prompting for multilingual researchers. Specifically, the highlight–popup–tooltip interaction successfully addressed the bottleneck of prompt reformulation,

yielding statistically significant reductions in task completion time across both tone-modification and translation tasks. However, while the implemented interface interventions accelerated text entry, they did not mitigate the cognitive burden of semantic verification required for cross-lingual tasks.

Consequently, these outcomes offered a new understanding of the context of use, user requirements, design, and evaluation. Regarding context of use and user requirements, the persistence of high cognitive load during translation suggested that efficiency requirements needed to be balanced with the need for trust and verification support. Future iterations were therefore required to address the gulf of evaluation and help users quickly validate accuracy. Moreover, design iterations needed to move beyond simple text replacement toward features that actively assisted in the validation process, thereby reducing the cognitive overhead identified in Assignment 5.

Finally, in addressing the limitations of the short-term study, we note that future evaluation efforts need to extend beyond isolated tasks to longitudinal studies that would better capture ecological validity. Such studies would allow us to observe how these tools integrate into extended, multi-application research workflows involving tools like Zotero and word processors. Insights from these longer-term deployments may, in turn, surface additional limitations and friction points that have not yet been identified or addressed.

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